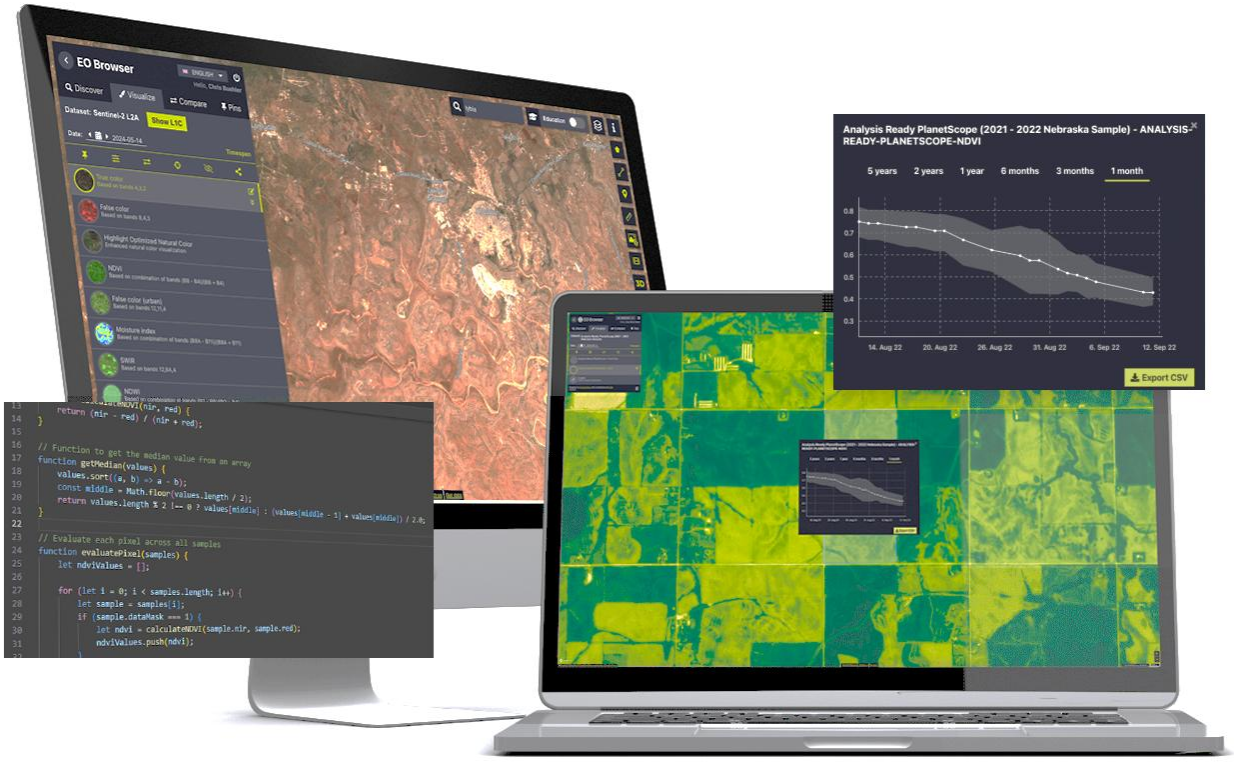
Technical Proposal For Development of Integrated Frameworks and Digital Solutions for FLR Monitoring, Evaluation, and Reporting under RECAP4NDC project



For

giz Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH

Submitted by

GISFY Building a smart world

GISFY Private Limited

DSL Abacus IT Park, Industrial Development Area Uppal,
Ranga Reddy Dist., Hyderabad, Telangana, 500039, India

Technical Submission Form

GISFY PRIVATE LIMITED
CIN: U74999UP2016PTC087694
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Hyderabad
30th April 2025
Tel No. +91 9971777963
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To,
The Head of Contracts and Procurement
GIZ GmbH, GDCO Office
New Delhi – 110 057

Subject: Proposal in response to Tender No. 83485307

Dear Sir/ Madam,

I/We the undersigned hereby offer to execute the scope of work and accordingly submit our offer in full compliance with terms & conditions of the bid.

The bid is being submitted as per the instructions mentioned in the tender documents.

Legal Name of Entity	GISFY Private Limited
Name of the Authorized Signatory	Reshma Parveen Shaik
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CIN Number/Registration Number	1. Uttar Pradesh: U74999UP2016PTC087694 2. Telangana: U74999TG2016PTC139570
Name of Directors with DIN/PAN	Reshma Parveen Shaik - 07621934 Azghar Hussain Mohammed - 07621930

For and on behalf of GISFY Private Limited.

Ms. Reshma Paveen Shaik
CEO, GISFY Private Limited

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How GISFY meets the technical requirements as per TOR

Fulfilment of technical-methodological concept requirements

S. No.	Technological-Methodological Concept	How we meet the requirement
1	Scalable and Modular System Architecture	We employ a microservices-based architecture, ensuring scalability, modular expansion, and interoperability with existing forest management systems.
2	Integration of Sentinel Hub APIs for Real-Time Satellite Imagery	Our team has extensive experience in integrating Sentinel Hub APIs for real-time satellite imagery retrieval, vegetation index analysis, and automated deforestation detection.
3	Development of Cross-Platform Mobile Applications	We develop mobile applications using Flutter for cross-platform (iOS & Android) compatibility, ensuring seamless field data collection with offline functionality.
4	Implementation of GIS-Based Decision Support System	We utilize OpenLayers, PostGIS, and GeoServer to build a GIS-based decision support system with interactive mapping and data analytics.
5	Real-Time Data Synchronization Between Mobile and Web Systems	We ensure seamless real-time data synchronization using Firebase and RESTful APIs, enabling immediate access to field data from web platforms.
6	User-Friendly UI/UX for Field and Administrative Users	Our UI/UX team follows human-centered design principles, ensuring intuitive navigation for both technical and non-technical users.
7	Role-Based Access Control and Data Security Measures	We implement secure authentication (OAuth 2.0, JWT) and data encryption to protect sensitive forest management data.
8	Automated Forest Cover Change Detection Using NDVI/NDWI	Our team specializes in remote sensing and machine learning models to automate NDVI/NDWI-based forest cover change detection and monitoring.
9	Cloud-Based Deployment for High Performance and Reliability	We deploy the system on AWS and Azure cloud platforms with auto-scaling, load balancing, and disaster recovery features.
10	Comprehensive Training and Knowledge Transfer Strategy	We provide structured training sessions, user manuals, and on-demand support to ensure successful adoption and sustainability of the platform.

Organizational Details

GISFY Private Limited is a is a globally recognized leader in digital transformation, geospatial intelligence, climate risk analysis, and conflict early warning systems. With a strong foundation in ICT system integration, data-driven decision-making, and AI-powered analytics, the organization specializes in developing scalable technology solutions for government agencies, international organizations, and humanitarian entities. Our expertise spans public sector digitalization, early warning platforms, disaster risk management, and predictive modeling, making us a trusted partner in enhancing resilience, security, and governance worldwide. We are recognized as a start-up by the Department of Industrial Promotion and Policy; and have worked with renowned organizations like Nanyang Technological University (Singapore), University of West Indies, Innovative Hydrology, and World Business Council for Sustainable Development (WBCSD), Belize Social Investment Fund, and various Government departments across the world.

We work across the world and provide services and solutions in the following areas of work.

- Agricultural and Forestry Applications Development
- Natural Resources Management
- GIS and ICT Platforms (web and mobile)
- Sentinel Hub API Integration
- Urban Development
- Hydromet and Conflict Early Warning Systems
- Geospatial Intelligence & GIS-Based Risk Mapping
- Disaster Risk Management & Climate Resilience
- Android Mobile App Development
- Geospatial mapping and Data Development
- Documentation & Knowledge Management

The following are further details and documentation proof of our organization

Item	Description
Name of the organization	GISFY PRIVATE LIMITED CIN No.: U74999UP2016PTC087694 PAN No.: AAGCG5903P GSTIN: 36AAGCG5903P1Z6
Type of the organization	Private Limited
Address	Ground Floor, DSL Abacus IT Park, Industrial Development Area, Uppal, Ranga Reddy District, Hyderabad, Telangana 500039, India;
Contact Details	Azghar Hussain Mohammed, Founder & CTO Tel: +91-9971777963 Email: azghar@gisfy.co.in; skype: yashikakki.

Management Structure

Key Personnel	Brief Profile
Dr. G. Sailu 	Dr. G. Sailu is a seasoned Project Manager with over 18 years of experience in leading complex, multi-stakeholder assignments in the domains of environmental management, forestry, and sustainable landscape restoration. He holds a Ph.D. in Environmental Science from Osmania University and an M.Sc. in Biotechnology from Periyar University. Dr. Sailu has successfully overseen the design and implementation of large-scale development projects funded by multilateral agencies, with a focus on planning, monitoring, team coordination, and client engagement. In his role as Project Manager for the RECAP4NDC assignment, Dr. Sailu will provide strategic leadership and full-cycle management oversight for the development of integrated Monitoring, Evaluation, and Reporting (MER) frameworks and associated digital tools for Forest Landscape Restoration (FLR). He will be responsible for managing work planning, expert coordination, stakeholder consultations, compliance with deliverables, and quality assurance across both Work Package 1 and Work Package 2. Dr. Sailu's experience includes managing interdisciplinary teams across diverse geographies, including India, Bhutan, and Kenya, ensuring timely delivery of project outputs while maintaining alignment with technical standards and institutional requirements. He brings a strong background in coordinating with national and state-level government institutions such as MoEFCC, FSI, and State Forest Departments, ensuring seamless integration of project activities with government priorities. He has led national-level assignments involving restoration planning, spatial data integration, and institutional capacity building. His ability to manage cross-functional teams including GIS experts, software developers, remote sensing analysts, and training specialists makes him well-suited to deliver the RECAP4NDC project objectives within the stipulated timeline and budget. Dr. Sailu will ensure adherence to the TOR-specified methodology, guide the implementation of stakeholder engagement and validation processes, and oversee the piloting of MER tools in Gujarat, Maharashtra, Uttarakhand, and Delhi NCR. His experience managing donor-funded projects aligns closely with GIZ's expectations on performance reporting, visibility, and adaptive planning.

	In addition to his project management responsibilities, Dr. Sailu will contribute to capacity-building efforts, coordinating the delivery of training modules, user manuals, and field implementation protocols. His leadership will ensure that the MER framework and digital solutions are not only technically sound but also practical, user-friendly, and sustainable for long-term institutional adoption. Dr. Sailu's proven project management track record, domain familiarity, and results-driven approach will be instrumental in delivering a high-impact, scalable MER system under the RECAP4NDC initiative.
Mr. Mohammad Moonis	Mr. Mohammad Moonis is a highly skilled Data Scientist and environmental analytics expert with a M. Phil. Natural Resource Management from the Indian Institute of Forest Management, Bhopal, and a M. Sc. Environmental Science from Jawaharlal Nehru University, New Delhi. With over 17 years of experience in data-driven environmental modeling, machine learning, and geospatial data analytics, Dr. Moonis brings a unique blend of domain expertise and advanced technical proficiency to support evidence-based forest management and landscape restoration. Dr. Moonis specializes in the application of statistical modeling, predictive analytics, and AI-based algorithms for environmental monitoring, land-use change analysis, and ecological risk assessment. His expertise will be critical to the development of intelligent features within the geospatial dashboard and mobile applications under the RCAP4NDC Gujarat project, ensuring real-time data interpretation, trend forecasting, and spatial decision support. Having worked extensively with international organizations such as FAO, UNEP, and GIZ, Dr. Moonis has led the design of machine learning models for deforestation prediction, carbon stock estimation, and afforestation impact evaluation. He is proficient in tools such as R, Python, Google Earth Engine, TensorFlow, and PostgreSQL/PostGIS, enabling him to build scalable, automated data pipelines and high-performing analytical dashboards. His work supports seamless integration of satellite imagery, mobile-based field data, and forest inventory datasets to produce actionable insights for planners and policymakers. Within this project, Dr. Moonis will be responsible for designing the data architecture and analytical models that underpin key functionalities, such as forest cover change detection (using NDVI/NDWI), patrolling pattern analysis, and spatial validation of working plan coupes. He will also lead the development of

	real-time alert systems and performance indicators that enhance the Department's ability to monitor restoration efforts and make data-backed interventions. Dr. Moonis has previously contributed to national-level decision-support tools and WebGIS platforms for forest degradation monitoring, agroforestry planning, and climate vulnerability mapping. His experience in combining structured and unstructured datasets, performing multi-temporal satellite image analysis, and building AI-enabled classification models will directly support the development of an intelligent and responsive geospatial system for the Forest and Environment Department, Government of Gujarat. In addition to his technical contributions, Dr. Moonis is a strong advocate of open data standards, reproducible research, and institutional capacity building. He will assist in creating user-friendly data visualization interfaces and analytical dashboards for use by field officers, GIS teams, and policy stakeholders. He will also lead sessions on data literacy and analytics interpretation as part of the project's knowledge management and training components. With a proven track record of using data science to drive conservation outcomes and policy transformation, Dr. Moonis will play a pivotal role in ensuring that the RCAP4NDC digital monitoring tools are robust, insightful, and future-ready—empowering Gujarat's Forest administration with the analytical capabilities needed for sustainable and adaptive forest landscape restoration.
Dr. Jayanthi Nagakumar 	Dr. Jayanthi Nagakumar is a distinguished Ecosystem Modeller and landscape ecologist with a Ph. D (Environmental Planning) from University of Mysore, and a M.Sc. (Botany) from University of Mysore. With over 36 years of experience in ecosystem simulation, spatial habitat modeling, and land degradation assessment, Dr. Jayanthi brings deep domain knowledge and advanced modelling expertise to support sustainable forest management and Forest Landscape Restoration (FLR) under climate change scenarios. Dr. Jayanthi has worked extensively at the interface of ecology, remote sensing, and spatial analytics, using data-driven models to simulate ecological processes, assess forest resilience, and support biodiversity conservation planning. Her contributions will be central to the RCAP4NDC Gujarat project, where she will develop ecosystem simulation models and restoration suitability maps, ensuring that mobile and dashboard tools are rooted in ecological intelligence and long-term environmental planning.

	With a career spanning multi-country projects in India, Dr. Jayanthi has applied ecosystem modeling techniques such as species distribution modeling (SDM), landscape connectivity analysis, and carbon sequestration forecasting to inform conservation strategies, afforestation programs, and natural resource management. She is proficient in platforms like InVEST, MaxEnt, R, GAMS, and ArcGIS ModelBuilder, allowing her to integrate ecological datasets with geospatial workflows for scenario analysis and restoration prioritization. As part of this project, Dr. Jayanthi will lead the design and implementation of spatial ecosystem models to assess forest health, land-use change dynamics, and restoration potential across Gujarat's forest landscapes. Her work will support the identification of ecologically sensitive zones, validate land cover classifications, and contribute to the development of decision-support layers within the geospatial dashboard enabling forest managers to visualize trade-offs and co-benefits of restoration actions. Dr. Jayanthi has previously contributed to the development of degradation vulnerability maps, ecosystem services quantification tools, and climate-resilient land-use strategies, making her expertise invaluable for building an integrated decision-making platform under the RCAP4NDC framework. Her work aligns with global standards and initiatives such as the Bonn Challenge, REDD+, and UNCCD Land Degradation Neutrality goals, ensuring policy and institutional relevance. In addition to his modelling work, Dr. Jayanthi is deeply involved in capacity building and applied research translation. She will support the creation of ecological training modules, conduct technical workshops for forest officers and GIS analysts, and guide the Forest Department in interpreting ecosystem indicators and model outputs for planning and monitoring purposes. With a rare combination of ecological insight, spatial modeling expertise, and hands-on experience in forestry and restoration contexts, Dr. Jayanthi will play a critical role in ensuring that the Gujarat Forest Monitoring platform is not only technologically advanced but also ecologically grounded and restoration-oriented contributing directly to Gujarat's vision of sustainable and climate-resilient forest landscapes.
Mr. Azghar Hussain Mohammed	Mr. Azghar Hussain Mohammed is a highly accomplished Software Engineer and enterprise systems architect with a Bachelor's degree in Computer Science Engineering from Kakatiya University, Andhra Pradesh, and over 22 years of experience in software development, platform architecture, and large-scale system integration across environmental, agricultural, and e-governance domains. His deep expertise in

full-stack development, secure API design, and high-performance digital platform engineering makes him exceptionally qualified to lead the software development components of the RCAP4NDC Gujarat Forest Monitoring Project.

Mr. Hussain has led the development of complex digital solutions for government and donor-funded projects, integrating WebGIS, mobile applications, and cloud infrastructure to address challenges in natural resource management, forestry, and climate adaptation. His core competencies include end-to-end software architecture, responsive mobile application design, backend service orchestration, and platform security compliance, with proficiency in technologies such as Microsoft ASP.NET, PHP, Python, Angular, React, and C#. He also brings strong experience in PostgreSQL, MongoDB, MySQL, and real-time data streaming from field-based IoT/GIS sources.

For this project, Mr. Hussain will play a central role in the design, development, and deployment of the mobile applications and the centralized geospatial dashboard, ensuring seamless functionality, real-time synchronization, and cross-platform scalability. His contributions will include building robust backend systems for forest cover change detection, patrol reporting, and coupe ground-truthing workflows ensuring that field data is processed securely and visualized instantly within the decision-support dashboard.

Mr. Hussain has previously developed cloud-based platforms for forest monitoring, seed supply chain tracking, e-marketplaces, and climate-vulnerable region assessments, supporting organizations such as the World Bank, UNDP, and state forest departments.

He is also adept at integrating geospatial processing tools (GeoServer, Leaflet, Mapbox) and real-time analytics modules, allowing the Gujarat Forest Department to monitor satellite data trends, generate reports, and respond to deforestation alerts with precision. His contributions will ensure that the digital tools are not only responsive and user-friendly but also compliant with data protection laws, ensuring secure handling of sensitive environmental and geolocation data.

Beyond development, Mr. Hussain plays a strong mentoring and support role facilitating developer team coordination, DevOps pipeline automation, and quality assurance testing to ensure agile delivery and high system reliability. He will also support capacity building by creating technical documentation,

	user training materials, and system maintenance protocols tailored to the needs of the Forest Department's IT teams. With a proven track record in delivering scalable and secure digital solutions for complex environmental systems, Mr. Azghar Hussain Mohammed will be instrumental in ensuring that the Gujarat RCAP4NDC platform is technically sound, fully interoperable, and capable of supporting real-time forest management and FLR monitoring at scale.
Mr. Durga Ajay Babu 	Mr. Ajay Babu is a highly skilled UI/UX designer with extensive experience in creating user-friendly and visually engaging digital interfaces for web and mobile applications. His expertise lies in user-centered design, accessibility, and interactive prototyping, ensuring seamless navigation and an intuitive user experience. He specializes in wireframing, prototyping, and front-end optimization, leveraging industry-standard tools such as Adobe XD, Figma, Sketch, and Axure RP. His technical proficiency in HTML, CSS, JavaScript, and Bootstrap enables him to develop pixel-perfect, responsive designs tailored to meet diverse user needs. He has played a crucial role in designing digital solutions for projects such as TIGRAM – Timber Traceability, Agri Entrepreneur Growth Foundation (AEGF), and NTR Smart Township, where he ensured optimal user experience, interactive visualization, and accessibility compliance. His work spans India, Africa, Cambodia, Nigeria, Belize, Vietnam, Singapore, Tajikistan, Kenya, Ethiopia, Bermuda, and the USA, reflecting his ability to adapt to global design trends and usability standards. His ability to translate complex user requirements into engaging, functional digital interfaces makes him an invaluable asset in enhancing the usability and effectiveness of digital platforms.
Mr. Pranay Kumar 	Mr. Pranay Kumar is an accomplished Geospatial Expert with a Master's Diploma in GIS & Remote Sensing from Jadavpur University, Kolkata, and a Master's in Geography from Patna University, Bihar. His strong academic background in geospatial science, cartography, and satellite image processing has provided him with the expertise to develop GIS-based solutions for environmental monitoring, land-use planning, and sustainable resource management. With an extensive understanding of spatial analysis, remote sensing applications, and geospatial data visualization, Pranay has played a pivotal role in integrating GIS technology with decision-support systems. His deep knowledge of cartographic principles, GIS modeling, and cloud-based geospatial data processing has allowed him to contribute effectively to complex GIS projects.

	With over 13 years of experience in GIS development, geospatial database management, and spatial decision-making, Pranay has worked on large-scale GIS implementations in India, Malawi, and Afghanistan. His expertise includes developing WebGIS applications, designing geospatial data infrastructure, and integrating remote sensing technologies for land-use analysis. He has contributed to various GIS-driven projects in forestry, agriculture, and disaster risk management, leveraging geospatial intelligence to support policy formulation and environmental sustainability initiatives. His work has been instrumental in building GIS portals, developing geospatial analytics models, and designing interactive mapping platforms for real-time decision-making. Pranay has been actively involved in conducting spatial risk assessments, mapping ecosystem vulnerabilities, and integrating climate adaptation strategies into GIS frameworks. His hands-on experience in WebGIS development, GIS database architecture, and cloud-based spatial data processing aligns seamlessly with this project's focus on forest monitoring, real-time satellite imagery integration, and mobile-based GIS applications. He is highly proficient in ArcGIS, QGIS, PostGIS, GeoServer, and OpenLayers, making him well-equipped to develop and deploy WebGIS applications for large-scale environmental projects. His expertise in Sentinel-2 and Landsat satellite image analysis enables him to extract valuable insights for forest cover change detection, vegetation health assessment, and afforestation monitoring. He has extensive experience in geospatial database design, spatial query optimization, and GIS-based predictive modeling, ensuring that the data-driven insights generated from GIS applications contribute effectively to conservation planning and forest management. His technical proficiency in developing spatial models, integrating AI-driven remote sensing analytics, and automating GIS workflows ensures that geospatial applications are scalable, efficient, and user-friendly. Throughout his career, Pranay has worked on numerous projects that utilize GIS technology for environmental monitoring, land degradation assessment, and sustainable forestry management. His work in developing vulnerability maps, land-use classification models, and geospatial dashboards for climate adaptation strategies demonstrates his ability to apply GIS in addressing environmental challenges. His experience in Sentinel Hub API integration for real-time remote sensing applications ensures that the forest monitoring platform

	will have highly accurate, up-to-date geospatial intelligence for conservation efforts. His expertise will be crucial in ensuring that the GIS system effectively supports data-driven decision-making, enabling policymakers and conservationists to assess afforestation efforts, detect deforestation trends, and optimize forest resource management strategies. With his strong background in geospatial science, GIS application development, and environmental analytics, Pranay will play a vital role in delivering a scalable, efficient, and impactful GIS solution for Gujarat's Forest conservation initiatives.
Mr. Kishan Kumar 	Mr. Kishan Kumar is an experienced full-stack web developer with over 8 years of expertise in developing scalable, high-performance web applications. His technical proficiency includes ASP.NET, PHP, Angular, ReactJS, Python, and JavaScript frameworks, along with extensive knowledge of database management systems such as MongoDB, PostgreSQL, and SQL Server. His expertise lies in developing dynamic, secure, and data-driven web platforms, integrating real-time analytics, cloud-based solutions, and robust API frameworks. He has contributed significantly to projects such as TGP, Lekha – Money Monitoring, and NTR Smart Township, where he played a key role in integrating secure payment gateways, WebGIS functionalities, and responsive data visualization tools. Having worked across India, Africa, Vietnam, Singapore, and the USA, Mr. Kishan has successfully developed and deployed enterprise-grade web solutions tailored to meet the needs of government agencies, private enterprises, and development organizations. His knowledge of agile methodologies, cloud security, and front-end optimization makes him a vital resource for modern, scalable, and high-performance web development.
Ms. Pankaja Chichagandi 	Ms. Pankaja Chichagandi is a seasoned full-stack developer with over 7 years of experience in software and web development, specializing in GIS-based applications, cloud computing, and fintech platforms. She has a strong background in C#, Java, PHP, Laravel, Drupal, and Python, coupled with expertise in GIS technologies such as ArcGIS, OpenLayers, and GeoServer. She has played a pivotal role in projects such as TIGRAM – Timber Management and Traceability, Agri Entrepreneur Growth Foundation (AEGF), and Van System, where she ensured seamless system integration, database optimization, and performance-driven application development. Her ability to

	develop intuitive, responsive, and scalable web solutions has contributed to the success of multiple data-driven projects in agriculture, forestry, and financial inclusion. Her experience across India, Africa, Singapore, and the USA highlights her adaptability to global technology trends and business requirements. With her expertise in web application security, cloud architecture, and API integrations, Ms. Pankaja is a key asset in delivering robust, secure, and innovative digital solutions.
Ms. Girija Priyadarsini Nayak 	Girija Priyadarshini Nayak is a highly proficient Web Developer with a Master's in Computer Applications (MCA) from Khallikote University, Odisha, India. With a strong academic foundation in software development, database management, and front-end/back-end integration, she has developed expertise in building scalable, secure, and high-performance web applications. Her education has equipped her with a deep understanding of programming languages, cloud-based deployments, and UI/UX design principles, ensuring that she can deliver seamless and interactive digital experiences. Her strong analytical skills, combined with her proficiency in modern web technologies, security protocols, and application optimization, make her a critical asset for complex IT solutions, especially for geospatial platforms and decision-support systems. With over three years of experience in web application development, UI/UX optimization, and database-driven solutions, Girija has successfully developed and deployed large-scale web platforms across various domains, including forest management, public health, environmental sustainability, and geospatial decision-making. She has played a crucial role in building dynamic, user-centric web applications that integrate real-time data analytics and geospatial visualization. Her experience extends to developing GIS-enabled dashboards, interactive mapping interfaces, and cloud-based application architectures, which align with this project's focus on forest monitoring, remote sensing, and decision-support tools. Girija's technical expertise spans multiple front-end and back-end technologies, including ASP.NET, ReactJS, JavaScript, jQuery, PostgreSQL, SQL Server, and cloud-hosted environments. She has extensive experience in geospatial web development, working with OpenLayers, Leaflet.js, GeoServer, and PostGIS, which are essential for building interactive GIS-based applications. Her proficiency in database management and API integration allows her to design efficient data pipelines for real-time information retrieval and visualization. She has

	implemented security best practices, including authentication protocols (OAuth 2.0), role-based access controls, and encryption techniques, ensuring that the applications she develops adhere to industry security standards. Additionally, her expertise in responsive web design and accessibility standards ensures that the GIS dashboards and geospatial data platforms are intuitive and user-friendly across multiple devices. She has been instrumental in developing WebGIS solutions for land-use monitoring, deforestation tracking, and afforestation planning, ensuring that policymakers and conservationists can make data-driven decisions based on real-time insights. Her work in integrating Sentinel Hub API services into WebGIS platforms enables seamless satellite imagery processing, vegetation index analysis, and land-cover classification. Her ability to optimize geospatial applications for real-time user interaction, data accuracy, and system responsiveness ensures that the web platform for forest monitoring will operate efficiently under large data loads and complex user interactions. Her role will be critical in ensuring that the web application provides seamless, user-friendly access to forest monitoring data, enabling decision-makers, field officers, and policymakers to effectively track afforestation efforts, patrol routes, and forest cover changes. With her expertise in scalable web development, GIS integration, and user-centric design, she will contribute to delivering a high-performance, secure, and interactive platform for forest conservation in Gujarat.
Mr. Nirmal K Shaji 	Mr. Nirmal K Shaji is a dedicated mobile application developer with over 4 years of experience in building high-performance, user-friendly mobile applications for Android and iOS platforms. He specializes in Flutter, Java, Kotlin, Ionic, and React Native, ensuring cross-platform compatibility, seamless UI/UX, and high-efficiency app performance. He has contributed to Engagement of an IT Agency for ICT Material Development (Agriculture), and Agri Entrepreneur Growth Foundation (AEGF), developing mobile solutions that integrate real-time data processing, offline functionality, and intuitive user interfaces. His expertise in mobile API integration, push notifications, and financial transaction security has been instrumental in enhancing user engagement and mobile accessibility. Having worked on mobile-based solutions in India, Africa, and the USA, he understands the nuances of developing scalable and secure applications that cater to diverse industry

	requirements. His technical skills in mobile architecture, real-time synchronization, and fintech integration make him a critical contributor to cutting-edge mobile technology solutions.
Mr. Mubasheer Ul Islam	Mr. Mubasher Ul Islam is a versatile mobile application developer with over 4 years of experience in Android and iOS development, specializing in mobile UI/UX optimization, high-performance app architecture, and data security. He has expertise in Java, Flutter, HTML, CSS, JavaScript, Python, and PHP, with a strong foundation in database integration using Room Database, SQL Server, and MongoDB. His contributions to Engagement of an IT Agency for ICT Material Development (Agriculture), and TIGRAM – Timber Management and Traceability, Van IT, Millet Market, Bharati Makka, and other application have been vital in delivering efficient, scalable, and secure mobile solutions. His ability to integrate GIS-based tracking, blockchain authentication, and fintech security into mobile applications ensures enhanced usability and seamless performance. With experience spanning India, Africa, and the USA, he has developed cross-industry technology solutions tailored to various mobile user experiences. His expertise in app security, performance optimization, and cloud-based mobile solutions makes him a key player in next-generation mobile application development.
Mr. Mangesh Subhash Nuel	Mangesh Subhash Neul is a highly skilled Quality Assurance (QA) Engineer with a Bachelor's degree in Electronics and Communication Engineering from Kavikulguru Institute of Technology and Science, Ramtek, Nagpur, Maharashtra, India. His strong academic foundation in computer systems, software development life cycle (SDLC), and automated testing frameworks has equipped him with an analytical mindset and deep technical expertise in ensuring software reliability, performance, and security. Mangesh has over three years of hands-on experience in software quality assurance, test automation, and mobile/web application testing. His professional journey has been focused on ensuring system reliability, performance efficiency, and security compliance for large-scale software applications. He has worked on multiple projects related to forestry, healthcare, environmental monitoring, and GIS-based solutions, performing comprehensive functional, regression, and performance testing to ensure high-quality, bug-free application deployments. His role has involved collaborating with development teams to identify vulnerabilities, conducting end-to-end test case execution, and implementing automated test

	scripts to enhance software robustness. His experience in testing GIS-based applications and WebGIS dashboards aligns perfectly with this project's focus on forest monitoring, geospatial data integration, and mobile-based field data collection. As a QA Engineer, Mangesh has developed a strong technical skill set in manual and automated testing, test planning, and system validation. He has expertise in functional testing, black box testing, boundary value analysis (BVA), and equivalence partitioning (ECP) to identify software defects efficiently. His proficiency in Selenium WebDriver, TestNG, and Jenkins enables him to implement automation test suites for continuous integration and deployment (CI/CD) workflows. He is experienced in mobile application testing, ensuring compatibility, performance, and usability of apps in various environments. His experience in performance and load testing helps optimize server response times and scalability, ensuring the WebGIS platform remains efficient and responsive. He has conducted extensive testing of geospatial decision-support tools, verifying the accuracy and reliability of GIS data visualization, map overlays, and spatial analytics. His experience in testing Sentinel Hub API integrations and real-time geospatial data processing ensures the platform's accuracy and responsiveness in forest monitoring applications. His bug-tracking methodologies, defect documentation, and test automation strategies contribute significantly to software stability and performance. He will conduct manual and automated testing for mobile data collection applications, WebGIS-based forest cover change detection systems, and decision-support tools for policymakers. His expertise in automated regression testing will help in identifying system vulnerabilities early, ensuring that the forest management platform operates with high efficiency and minimal downtime. His contributions will be instrumental in delivering a robust, error-free, and high-performance system for real-time forest monitoring and conservation efforts in Gujarat.
Ms. Fathima Begum	Fathima Begum is a highly skilled Deployment Engineer with a Bachelor's degree in Computer Science Engineering from Vignana Bharathi Institute of Technology, JNTU Hyderabad, India. With a strong academic foundation in computer systems, cloud computing, and network security, she has developed deep expertise in deploying, managing, and optimizing cloud-based infrastructure solutions. Her technical education has provided her with a robust understanding of database management, IT security, automation, and scalable application

hosting, all of which are crucial in modern cloud computing environments.

With over three years of professional experience, Fathima has been instrumental in managing IT infrastructure, deploying cloud applications, and optimizing server performance for various large-scale projects. She has played a key role in configuring, deploying, and maintaining cloud-based applications in mission-critical environments, ensuring high availability, security, and scalability. Her work spans multiple domains, including forestry, health monitoring systems, GIS-based environmental platforms, and government-backed digital transformation initiatives. She has successfully led server migration projects, optimized system performance through cloud automation, and implemented high-security access controls for various organizations. Her experience extends to working with geospatial applications, where she has managed the deployment of GIS-based forest monitoring tools, real-time environmental data processing systems, and secure remote sensing applications.

Fathima possesses a diverse set of technical expertise and specialized skills in cloud infrastructure management, IT security, automation, and DevOps workflows. She is well-versed in AWS Cloud Services, including EC2, RDS, S3, IAM, CloudFront, ELB (Elastic Load Balancing), and CloudFormation, making her adept at deploying and maintaining scalable cloud environments. She is proficient in server management and optimization, ensuring high system reliability and efficiency by actively monitoring Ubuntu and Windows servers. Her experience in automated deployment and DevOps tools includes working with CI/CD pipelines, Jenkins, and infrastructure as code (IaC) using CloudFormation and Terraform. She has extensive experience in data backup and disaster recovery planning, including AWS S3 backup management, multi-region RDS instance replication, and high-availability cloud architecture setup.

Throughout her career, Fathima has worked on several large-scale projects, including the deployment of geospatial applications for environmental conservation, GIS-based decision support systems for forestry, and digital platforms for real-time monitoring of natural resources. She has been actively involved in configuring and deploying WebGIS platforms, ensuring seamless integration with Sentinel Hub APIs for real-time satellite data processing. Her work in cloud-based GIS hosting has facilitated the visualization and analysis of environmental data, contributing to forest management,

	afforestation planning, and land-use monitoring. By leveraging her expertise in cloud security, role-based access controls, and encrypted data transmission, she has ensured that these platforms comply with data privacy regulations and cybersecurity best practices. She will be responsible for optimizing AWS cloud infrastructure to support real-time geospatial data processing, integrating Sentinel-2 satellite imagery for forest monitoring, and ensuring the system's high availability and security.
Mr. Siddhartha Goli 	Mr. Siddhartha Goli is a meticulous documentation specialist with expertise in technical writing, compliance documentation, and process standardization. He has been responsible for preparing user manuals, administrative guides, inception reports, and training documentation for complex software systems and technology-driven projects. His work on TIGRAM – Timber Traceability, Agri Entrepreneur Growth Foundation (AEGF), and Van IT has been instrumental in ensuring well-structured, high-quality documentation that enhances knowledge transfer and project transparency. His strong proficiency in Microsoft Office, Advanced Excel, and Adobe Photoshop enables him to create detailed, visually engaging, and easy-to-understand documentation. Having worked in India, he has provided documentation support for various technology-driven initiatives, ensuring that project deliverables are well-documented, standardized, and easily comprehensible for all stakeholders. His role in training, compliance, and knowledge management significantly enhances project efficiency and usability.

1. Technical-Methodological Design

1.1 Strategy

1.1.1 Interpretation of the objectives in the TORs, critical examination of tasks

The RECAP4NDC Project outlined in the Terms of Reference (TOR) aims to develop an integrated Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) initiatives in selected landscapes across India. The central objective is to design and pilot scalable, participatory, and evidence-based MER frameworks, along with digital tools, that can support forest restoration planning and reporting at state and national levels. This will be accomplished through co-development of standardized MER indicators, methods, and tools; application of modern technologies such as remote sensing, GIS, drones, IoT, and AI; and stakeholder-driven institutional integration.

A key focus of the project is on creating MER systems that are aligned with India's national priorities and international commitments, such as the Bonn Challenge, UN SDGs, Paris Agreement, State of Forest Reports, and National Forest Inventory. By designing standardized frameworks and protocols, the project seeks to enable consistent, comparable, and outcome-focused monitoring across diverse restoration contexts. Additionally, the assignment requires integration with existing national and sub-national forest governance systems to ensure uptake and sustainability.

Another significant goal is the development and deployment of digital tools including mobile applications, dashboards, and GIS platforms to collect, visualize, and analyze restoration data. These tools should support real-time data collection, remote validation, and interactive reporting for various stakeholders, from field-level practitioners to state and central policymakers. The digital solutions must be user-friendly, flexible for use in different landscapes, and interoperable with existing systems, enabling seamless data exchange and institutional adoption.

Furthermore, the assignment emphasizes piloting the MER framework and tools in selected landscapes across Gujarat, Maharashtra, Uttarakhand, and Delhi NCR. Field testing and refinement will be critical to ensure that the systems are functional, scalable, and context-appropriate. The pilots will demonstrate how MER frameworks and digital tools can track ecological outcomes (e.g., forest cover, soil health), social outcomes (e.g., livelihoods, participation), and institutional outcomes (e.g., policy alignment, stakeholder coordination) of restoration interventions.

The assignment also underscores the need for participatory development of all outputs. This includes engaging key national and sub-national stakeholders such as MoEFCC, FSI, SFDs, IUCN, GIZ, and restoration practitioners through interviews, consultations, and workshops. This participatory approach ensures contextual relevance, capacity building, and stakeholder ownership of the MER systems and tools.

In addition, the project calls for strong knowledge management and capacity-building components. This includes developing user manuals, SOPs, and training modules for field staff and government officials. Capacity-building efforts such as training-of-trainers programs and regional workshops will be necessary to build long-term institutional capacity to use and manage the MER systems.

The assignment has the following specific objectives:

- Design of integrated and standardized Monitoring, Evaluation, and Reporting (MER) frameworks for Forest Landscape Restoration (FLR).
- Co-development of restoration indicators, methods, and tools aligned with national and global goals.
- Design and deployment of digital tools such as mobile apps, dashboards, and GIS platforms for MER.
- Field testing and piloting of the MER framework and tools in selected FLR landscapes.
- Stakeholder engagement and participatory development of MER systems.
- Capacity building and knowledge management for sustainable institutional adoption.
- Integration of Sentinel Hub APIs for High-Resolution Satellite Imagery Processing.

1.1.2 Contextualization for Gujarat State

The proposed solution is specifically designed to meet the needs of the Gujarat State Rural Livelihood Mission (GSRLM) under the Government of Gujarat. The modules including Grievance Redressal, Geo-tagging, and Knowledge Hub are tailored to support Self-Help Groups (SHGs), Village Organizations (VOs), and Cluster-Level Federations (CLFs) operating in the state's diverse rural landscape. Special attention will be given to incorporating Gujarati language support, integrating with existing Gujarat-specific MIS systems, and ensuring usability for rural women and functionaries with varying levels of digital literacy.

1.1.3 Description and Justification of the contractors strategy for delivering the services put out of tender

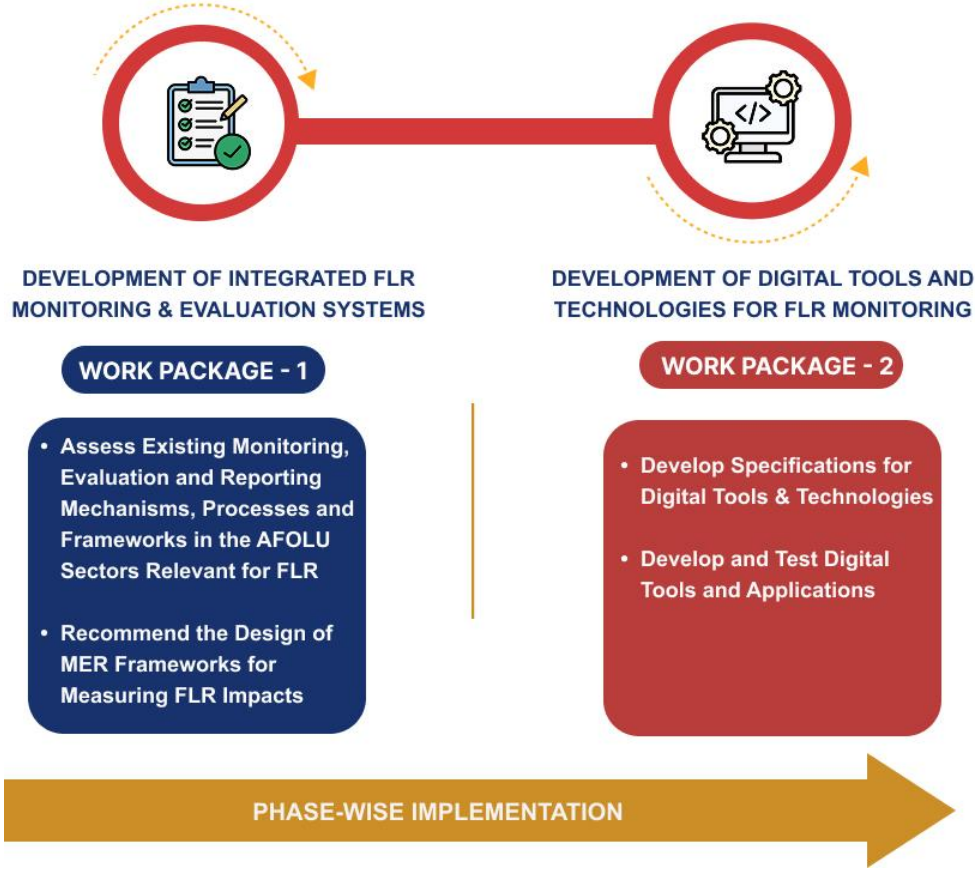
The consortium led by GISFY Private Limited, with Everain Global Services Private Limited as a strategic partner, proposes a rigorous, participatory, and technology-enabled approach to deliver the services under the RECAP4NDC assignment. The strategy is anchored in the consortium’s deep experience in ecological monitoring, geospatial system development, digital MER tools, and institutional engagement.

The overall implementation strategy is designed to support the development of a robust Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) in India. This includes both the technical development of frameworks and digital tools, and the human and institutional capacity-building required to ensure long-term adoption and sustainability.

We structure our delivery around the two Work Packages specified in the TOR:

- Work Package 1: Development of an integrated FLR monitoring and evaluation system in India.
- Work Package 2: Development of digital tools and technologies to operationalize the MER system and its institutional adoption.

Each Work Package consists of targeted activities that are mutually reinforcing and responsive to the TOR's requirements.



Work Package 1: Development of Integrated FLR Monitoring & Evaluation Systems

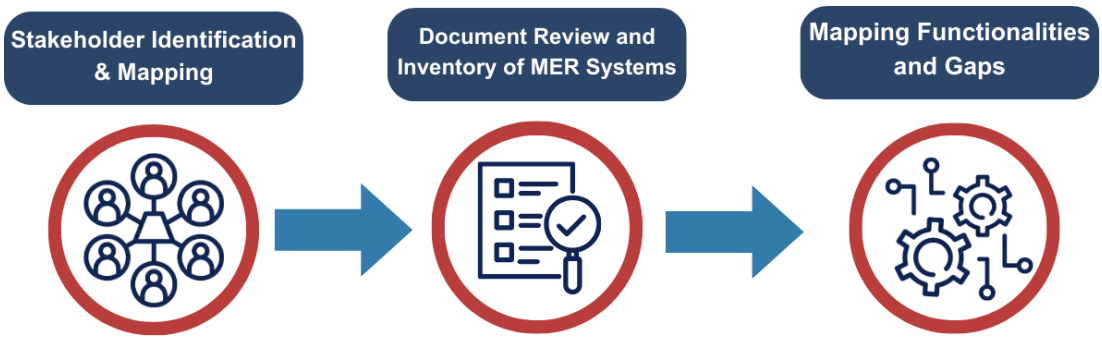
This work package aims to create a robust and integrated Monitoring, Evaluation, and Reporting (MER) system for Forest Landscape Restoration (FLR) across India. It focuses on building upon existing MER frameworks within the Agriculture, Forestry, and Other Land Use (AFOLU) sectors and tailoring them to monitor FLR outcomes. This work will support national commitments under the Bonn Challenge, Land Degradation Neutrality targets and NDC goals.

Activity 1.1: Assess Existing Monitoring, Evaluation and Reporting Mechanisms, Processes and Frameworks in the AFOLU Sectors Relevant for FLR

This activity will systematically analyze current MER systems in place across AFOLU sectors to evaluate their alignment with FLR indicators and outcomes. The goal is to identify strengths, weaknesses, and opportunities for harmonization to ensure a cohesive FLR monitoring structure across India.

Task 1.1.1: Stakeholder Mapping and Inventory of Existing MER Frameworks

This task involves a comprehensive review of stakeholders and existing MER frameworks.



Stakeholder Identification and Mapping

- Identify key stakeholders from national, state, and district levels across AFOLU sectors. This includes ministries (MoEFCC, MoA&FW), technical agencies (FSI, NRSC), and NGOs involved in restoration.
- Map institutional mandates and roles related to data collection, monitoring, and reporting, focusing on overlaps and gaps relevant to FLR.
- Engage stakeholders through interviews, focus group discussions, and participatory workshops to document their current and potential roles in FLR monitoring.
- Evaluate communication channels and coordination structures between institutions involved in MER.
- Utilize GIS to visualize spatial presence and operational zones of stakeholders.
- Document the historical engagement of each stakeholder in MER systems and FLR initiatives.
- Analyze capacity levels, funding, and technical expertise available with stakeholders.
- Categorize stakeholders by influence, interest, and institutional readiness for collaboration in the proposed FLR monitoring system.

Document Review and Inventory of MER Systems

- Collect policy guidelines, manuals, annual reports, technical standards, and existing toolkits related to MER in forestry, agriculture, and watershed programs.
- Analyze spatial and temporal coverage of monitoring systems, including integration with geospatial platforms such as Bhuvan, VEDAS, and SEPAL.
- Examine indicators, data standards, reporting intervals, and compliance mechanisms adopted under each MER framework.
- Assess the presence of monitoring technologies like mobile apps, drones, and satellite imagery within these systems.
- Identify reporting requirements for national and international commitments (e.g., SDGs, UNCCD, UNFCCC, CBD).
- Create a repository of datasets, including metadata such as spatial resolution, periodicity, and responsible agency.
- Document interoperability challenges among existing digital MER platforms.
- Highlight best practices and case studies that can inform the design of an FLR-centric MER framework.

Mapping Functionalities and Gaps

- Develop a comparative matrix of key functionalities across MER systems, evaluating efficiency, inclusiveness, and scalability.
- Identify data collection duplication, inconsistent methodologies, and fragmented reporting lines that hinder effective FLR monitoring.
- Examine institutional silos that impede coordination and data sharing across ministries and departments.
- Analyze technological and infrastructure barriers to real-time monitoring, particularly in remote and tribal areas.
- Assess user-friendliness and field applicability of existing MER tools, identifying needs

for capacity building.

- Review institutional incentives and mandates that may impact data accuracy or compliance with MER procedures.
- Highlight areas where frameworks meet international standards and where localization is needed.
- Recommend areas for consolidation, integration, or reengineering of current systems to improve alignment with FLR objectives.

Task 1.1.2: Analytical Review of MER Frameworks and Institutional Capacities

This task involves a detailed analysis of institutional capacities and performance indicators.

Institutional Capacity Assessment

- Conduct a detailed analysis of institutional roles, mandates, and capacities for MER functions relevant to FLR.
- Evaluate the availability of technical personnel, digital infrastructure, and logistics at national and state levels.
- Identify capacity gaps in terms of knowledge, skill sets, and resource allocation among implementing agencies.
- Review previous MER training programs and assess their applicability to FLR-specific needs.
- Analyze coordination structures and governance challenges affecting institutional performance in MER.
- Examine budget allocations and funding mechanisms supporting MER activities across departments.
- Determine the potential for digital transformation and automation in MER functions.
- Propose a capacity development roadmap for scaling FLR monitoring through institutional strengthening.

Performance Indicator Analysis

- Compile and categorize indicators currently used in forestry, watershed, and agriculture programs.
- Evaluate indicator sensitivity to restoration outcomes like biomass increase, biodiversity enhancement, and land degradation neutrality.
- Assess consistency in definitions, baseline measurements, and periodicity across departments.
- Determine the extent of disaggregated data availability (gender, geography, livelihoods).

- Examine whether indicators are outcome-focused or activity-centric, and their alignment with global standards.
- Identify indicators that can serve as proxies for ecosystem services and resilience.
- Recommend new indicators for FLR integration based on scientific validity and field practicality.
- Propose standardized templates for indicator tracking, reporting, and impact assessment.

Global Benchmarking

- Review internationally recognized MER systems, including CIFOR's FLR monitoring guidelines and FAO's SEPAL platform.
- Identify technological, operational, and institutional features that have proven successful in other regions.
- Evaluate adaptability of these systems to Indian federal structures, data needs, and program cycles.
- Compare open-source versus proprietary platforms for scalability and cost-effectiveness.
- Investigate policy and incentive frameworks that support MER adoption in other countries.
- Draw lessons from community monitoring initiatives and citizen science models globally.
- Develop a benchmark matrix comparing Indian systems with global counterparts.
- Formulate recommendations for integrating best practices into India's proposed FLR MER system.

Task 1.1.3: Synthesis and Development of a Gap Analysis Report

This task involves synthesizing findings and developing a gap analysis.

Draft Gap Analysis Report

- Synthesize insights from the Stakeholder Mapping, Inventory of existing MER frameworks, and the analytical review of MER frameworks and institutional structures into a comprehensive narrative that outlines the current status and effectiveness of existing MER frameworks.
- Identify key gaps in data collection, institutional mandates, technological infrastructure, and interdepartmental coordination.
- Provide a thematic breakdown of gaps by sector forestry, agriculture, land, and water and by geographic zone.

- Structure the report with visual elements like graphs, matrices, and infographics to enhance understanding.
- Outline strategic areas for harmonization with national environmental and climate-related reporting platforms.
- Integrate case studies from previous successful restoration monitoring initiatives in India and abroad.
- Ensure that the report addresses both operational and strategic dimensions of the MER ecosystem.
- Position the report as a foundational document to guide the design of the FLR-specific MER framework in subsequent activities.

Stakeholder Validation

- Design a participatory validation process engaging representatives from key ministries, line departments, and civil society.
- Organize regional consultations and national workshops to review and validate the findings of the gap analysis report.
- Use interactive methods such as SWOT analysis, problem tree analysis, and participatory ranking tools.
- Capture stakeholder inputs through surveys, group discussions, and plenary feedback sessions.
- Address diverse perspectives, especially from local implementation agencies and community-based organizations.
- Document all feedback with attribution to inform a transparent revision process.
- Revise the draft gap analysis report based on inputs received and prepare a final validated version.
- Ensure that the validated report is accepted and endorsed by key nodal agencies for official integration into planning.

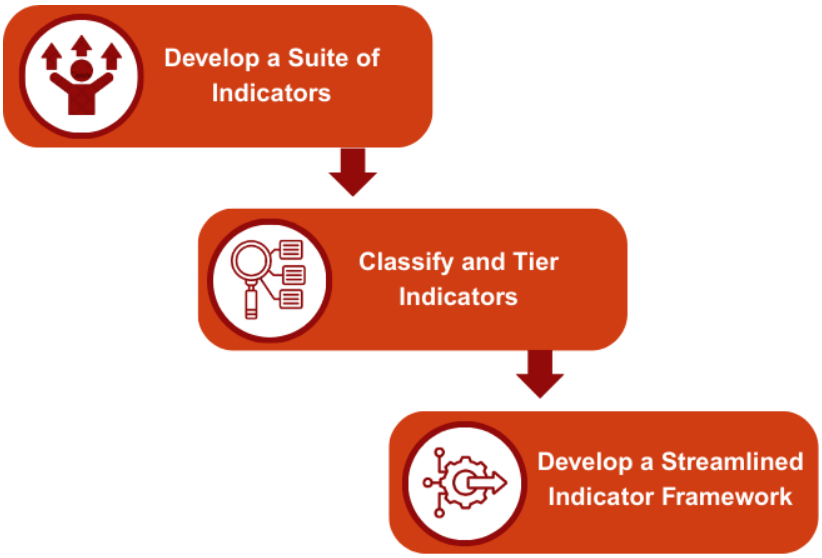
Final Synthesis and Roadmap

- Develop a strategic roadmap that outlines short-term, medium-term, and long-term interventions for bridging identified gaps.
- Assign responsibilities to institutional actors for each phase of implementation based on their mandates and capacities.
- Propose timelines, budgets, and indicators for monitoring progress on roadmap implementation.
- Integrate capacity-building and technology adoption strategies into the roadmap.
- Align the roadmap with existing policy frameworks such as the National Action Plan on Climate Change (NAPCC), NDCs, and SDG 15.3 targets.
- Highlight potential areas for donor and multilateral engagement to support implementation.
- Include contingency planning for operational risks, funding limitations, or stakeholder resistance.
- Present the roadmap in a modular format to allow phased implementation and customization by state governments.

Activity 1.2: Recommend the Design of MER Frameworks for Measuring FLR Impacts

This activity focuses on designing new and improved MER frameworks tailored for FLR that build upon the findings from Activity 1.1. It will propose specific indicators, data workflows, governance mechanisms, and reporting formats necessary to measure the ecological, social, and economic outcomes of FLR efforts in India.

Task 1.2.1: Define FLR-specific Indicators and Metrics



Develop a Suite of Indicators

- Identify biophysical indicators such as forest cover, biomass, soil carbon, land productivity, and water availability to measure ecosystem restoration.
- Include socio-economic indicators such as household income diversification, employment in restoration activities, and gender inclusion metrics.
- Consider institutional indicators like governance effectiveness, inter-agency collaboration, and policy alignment.
- Ensure indicators are scalable and usable at national, state, and landscape levels.
- Validate indicator selection with stakeholders to ensure relevance and measurability.
- Align indicators with SDGs (particularly 15.3), Bonn Challenge Barometer, and India's NDC framework.
- Propose core vs. optional indicators based on implementation contexts.
- Ensure indicators support adaptive management and decision-making.

Classify and Tier Indicators

- Establish a tiered structure with national-level, state-level & landscape-level indicators.
- Specify data collection methods (field measurements, remote sensing, etc.) for each tier.
- Outline how local data will be aggregated to inform national reporting.
- Define spatial and temporal scales for measurement and reporting.
- Clarify roles of different stakeholders in data collection and aggregation.
- Allow flexibility for states to include additional indicators relevant to their context.

Develop a Streamlined Indicator Framework

- Consolidate a comprehensive list of potential indicators into a core set for regular monitoring.
- Prioritize indicators that are cost-effective, easy to measure, and policy-relevant.
- Ensure the framework aligns with national and international reporting requirements.
- Incorporate feedback loops for periodic review and revision of the indicator set.
- Document the scientific basis and rationale for selecting each indicator.

Task 1.2.2: Design Data Collection Methodologies and Protocols

This task includes the following sub-tasks:

1. Develop Standardized Protocols
   - Develop detailed, step-by-step protocols for data collection, including sampling design, measurement techniques, and frequency.
   - Incorporate quality assurance and quality control (QA/QC) mechanisms to ensure data accuracy and reliability.
   - Outline standard operating procedures (SOPs) for data entry, storage & management.
   - Ensure protocols are adaptable to diverse field conditions and resource availability.

2. Select Appropriate Data Collection Technologies
   - Evaluate the use of remote sensing, drones, mobile apps, IoT devices, and other technologies for data collection.
   - Assess the feasibility and cost-effectiveness of different technologies in the Indian context.
   - Provide guidance on technology selection based on the specific indicator, scale of measurement, and available expertise.

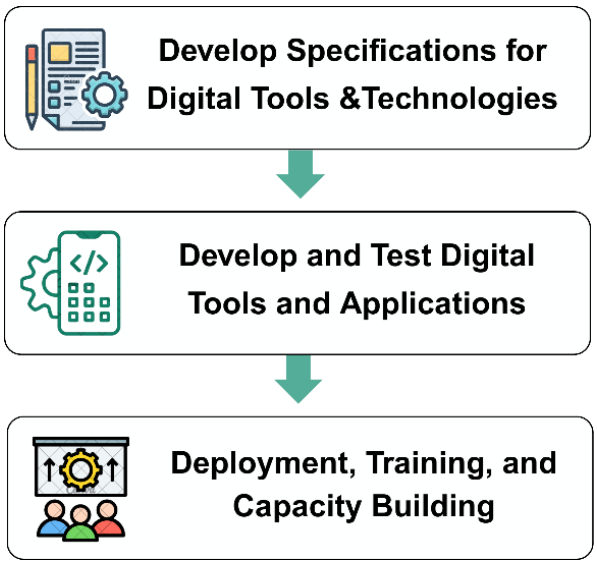
3. Design a Scalable Data Management System
   - Develop a centralized database architecture for storing, managing, and analyzing FLR data.
   - Ensure data security, integrity, and accessibility for authorized users.
   - Establish data sharing agreements and protocols among different stakeholders.
   - Enable interoperability with existing national and state-level data platforms.

Task 1.2.3: Outline Institutional Arrangements and Governance Mechanisms

1. Define Roles and Responsibilities
   - Clearly define the roles and responsibilities of different government agencies, research institutions, NGOs, and local communities in the MER process.
   - Establish mechanisms for inter-agency coordination and collaboration.
   - Define data sharing agreements and protocols to facilitate information exchange.
2. Establish Governance Structures
   - Propose a multi-stakeholder governance framework for overseeing the implementation of the FLR MER system.
   - Outline decision-making processes, including mechanisms for stakeholder input and conflict resolution.
   - Ensure accountability and transparency in data collection, analysis, and reporting.
3. Develop Feedback and Adaptive Management Mechanisms
   - Design mechanisms for data feedback and dissemination to inform policy and practice.
   - Incorporate adaptive management principles into the MER system to allow for adjustments based on new information and changing conditions.
   - Ensure that the MER system supports continuous improvement in FLR implementation.

Work Package 2: Development of Digital Tools and Technologies for FLR Monitoring

This work package focuses on developing the digital tools necessary to support the implementation of the FLR monitoring framework designed in Work Package 1. This includes creating software applications, data dashboards, and decision support systems to facilitate data collection, analysis, visualization, and reporting.



Activity 2.1: Develop Specifications for Digital Tools and Technologies

This activity involves defining the technical requirements, functionalities, and user needs for the digital tools to be developed.

Task 2.1.1: Conduct Detailed Needs Assessment

1. User Requirements Elicitation
   - Conduct workshops and interviews with State Forest Departments (SFDs), Forest Survey of India (FSI), researchers, and community members.
   - Document specific user needs related to data collection (e.g., field data entry, integration with existing devices), analysis (e.g., statistical functions, spatial analysis), visualization (e.g., maps, charts), and reporting (e.g., automated report generation, compliance reporting).
   - Prioritize requirements based on user feedback, alignment with FLR objectives, and technical feasibility.
   - Develop user personas to represent different user groups and their specific needs.
   - Establish a mechanism for ongoing user feedback throughout the development process.

2. Functional Requirements Specification
   - Translate user requirements into detailed functional specifications.
   - Define data inputs (e.g., remote sensing data, field data, socio-economic data), processing logic (e.g., algorithms for biomass estimation, change detection), outputs (e.g., reports, alerts, maps), and system behavior.
   - Specify the required functionalities of each digital tool, including data capture, storage, retrieval, processing, analysis, and presentation.
   - Define data validation rules, error handling procedures & system security requirements.
   - Specify algorithms, models, and analytical methods to be incorporated (e.g., time series analysis, spatial statistics).

3. Technical Requirements Specification
   - Define the technical architecture, hardware, software, and infrastructure requirements.
   - Specify data storage requirements (e.g., database type, capacity), security requirements (e.g., access control, encryption), scalability requirements (e.g., ability to handle increasing data volumes and users), and interoperability requirements (e.g., ability to exchange data with other systems).
   - Select appropriate technologies, programming languages, and development platforms (considering factors like cost, performance, and maintainability).
   - Define system performance criteria (e.g., response time, throughput, availability).
   - Address data protection and privacy considerations in compliance with relevant regulations.

Task 2.1.2: Design System Architecture and Data Flows

- Design the overall system architecture, including the relationships between different components and modules.
- Select appropriate architectural patterns and frameworks (e.g., microservices, cloud-based architecture).
- Ensure the system is modular, scalable, and maintainable.

- Define the interfaces between different system components.
- Specify the hardware and software components required for the system.

1. Physical System Architecture

This section outlines the physical infrastructure for the FLR Monitoring and Evaluation System, including server setup, user access mechanisms, and connectivity modes. It is designed to support seamless data collection, processing, and visualization across national and state-level institutions, field officers, and technical agencies. The users are divided into the following categories:

- **Field Staff (RFOs, Forest Guards, Patrolling Teams):** Use mobile application with offline support for patrolling, plantation validation, and ground-truthing. Upload GPS-tagged observations, notes, and media files. Data is synced automatically when internet connectivity is restored.
- **MoEFCC & National FLR Authorities:** Access the national FLR Dashboard and WebGIS via secure web portals to monitor progress toward international commitments (Bonn Challenge, SDG 15.3, NDCs). Utilize summary reports, interactive maps, and indicators for national reporting and strategy planning.
- **State Forest Departments (SFDs):** Use state-level dashboard access to coordinate field-level monitoring, review ground-truthing data, and generate state-wise FLR reports. Oversee technical workflows and ensure integration of state initiatives into the national MER framework.
- **GIS Analysts and Remote Sensing Experts:** Access and process satellite datasets (e.g., Sentinel-2, Landsat) for NDVI, NDWI, and forest cover analytics. Update geospatial layers via secure logins and contribute processed outputs to WebGIS and indicator dashboards.
- **Technical Partners and Research Institutions:** Access backend datasets, indicators, and validation tools for framework development, peer-review, and scientific analysis. May also contribute to refining methodologies and assist in field data interpretation.
- **GIZ Project Managers and Implementation Partners:** Access administrative dashboards to monitor deliverables, quality metrics, training rollouts, and compliance to milestones. Review intermediate outputs and ensure platform development aligns with project objectives.
- **System Administrators:** Configure and maintain both server infrastructure and user access roles. Oversee system uptime, data backups, security layers, and patch management. Support data standardization and ensure integration with external platforms (e.g., Bhuvan, VEDAS). Administer authentication controls & manage APIs.

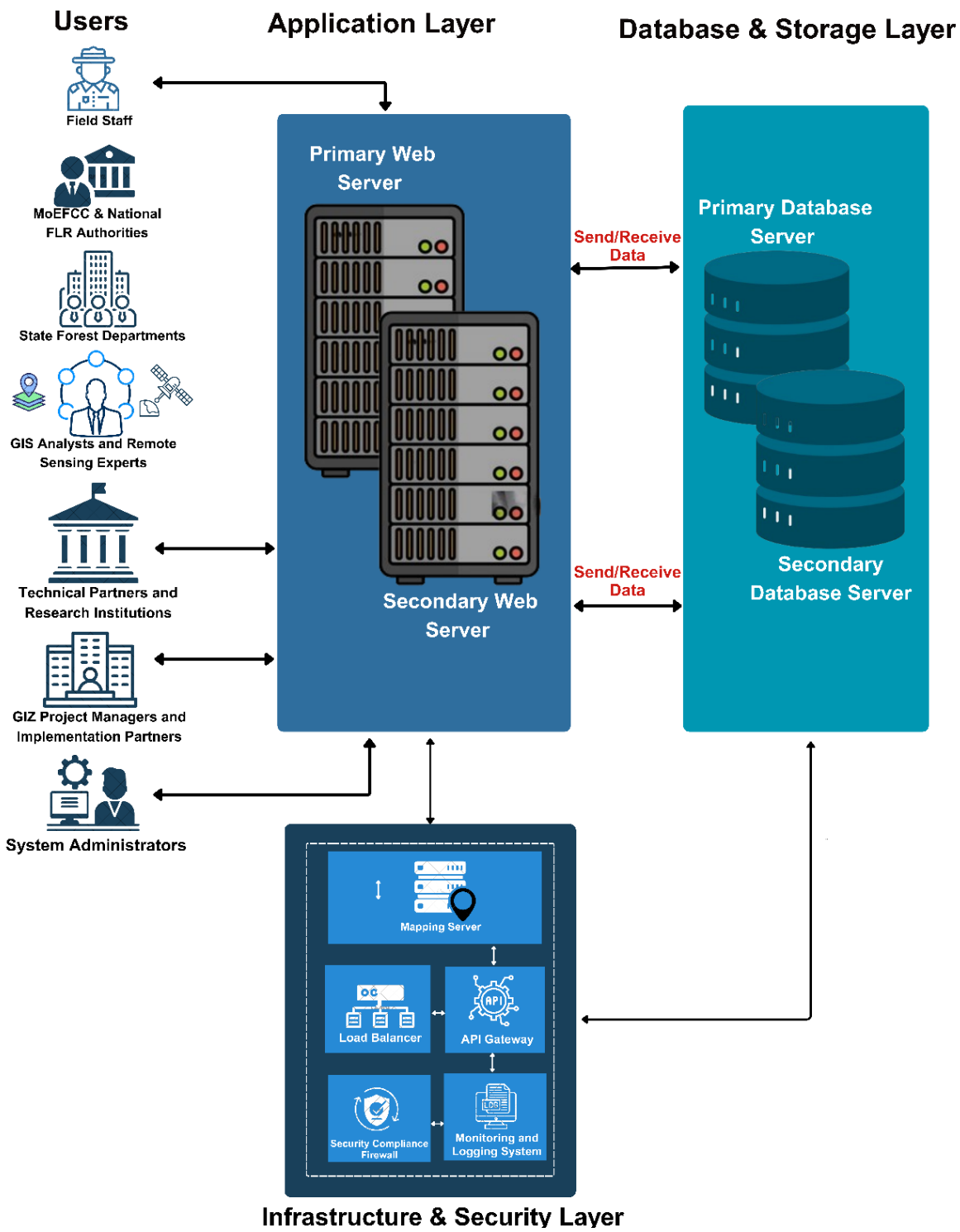
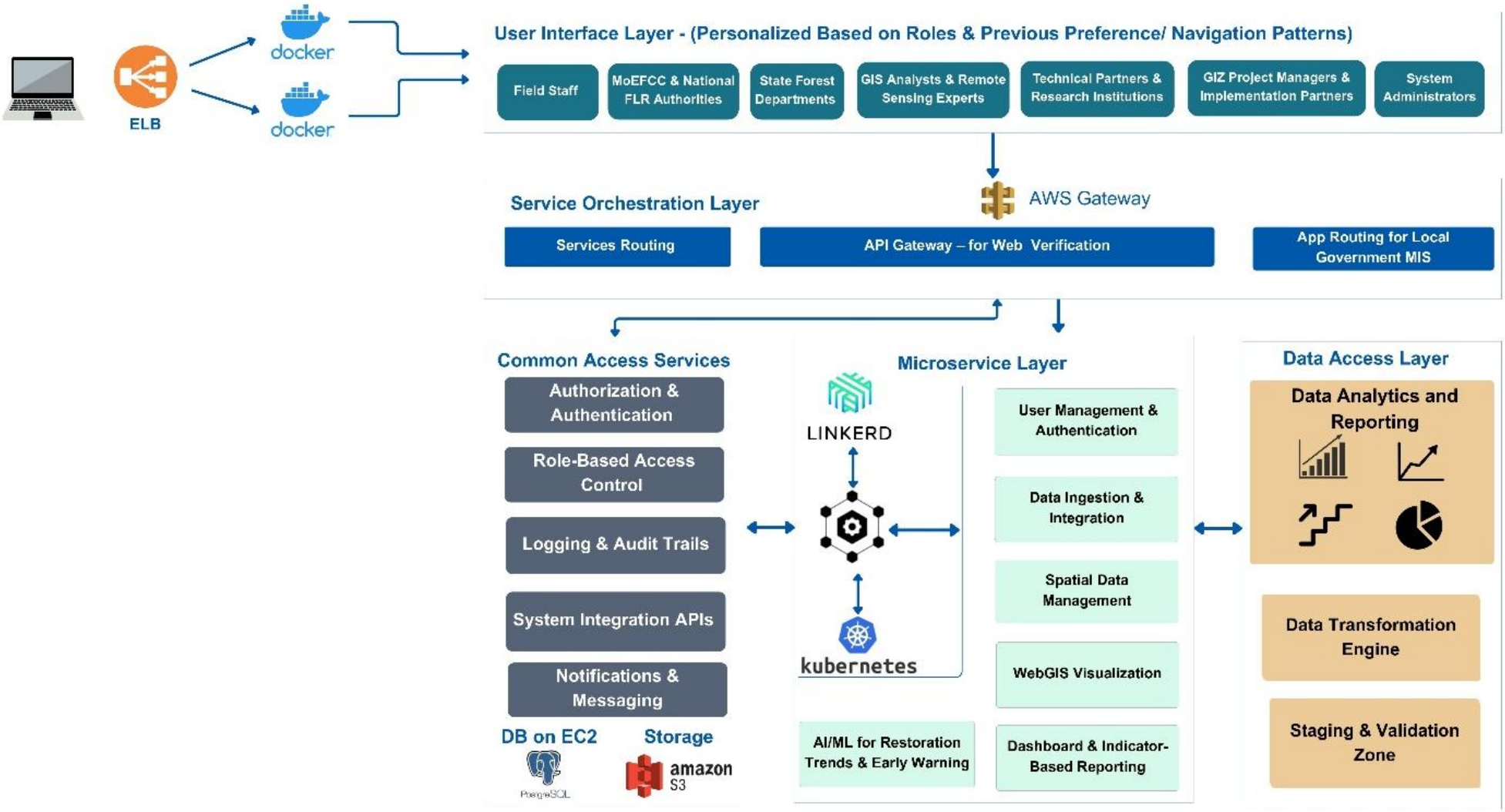


FIGURE 1: PHYSICAL ARCHITECTURE

2. Logical Architecture

The logical system architecture of the FLR Monitoring and Evaluation (MER) platform will be designed to ensure seamless interoperability, real-time monitoring, robust analytics, and scalable functionality. The architecture is modular and layered for clarity, security, and performance, and is structured into four main layers:

The tentative logical architecture consists of the following components as given below.


ELB
docker
docker
User Interface Layer - (Personalized Based on Roles & Previous Preference/ Navigation Patterns)
Field Staff
MoEFCC & National FLR Authorities
State Forest Departments
GIS Analysts & Remote Sensing Experts
Technical Partners & Research Institutions
GIZ Project Managers & Implementation Partners
System Administrators
Service Orchestration Layer
AWS Gateway
Services Routing
API Gateway – for Web Verification
App Routing for Local Government MIS
Common Access Services
Authorization & Authentication
Role-Based Access Control
Logging & Audit Trails
System Integration APIs
Notifications & Messaging
DB on EC2
PostgreSQL
Storage
amazon S3
Microservice Layer
LINKERD
kubernetes
User Management & Authentication
Data Ingestion & Integration
Spatial Data Management
WebGIS Visualization
AI/ML for Restoration Trends & Early Warning
Dashboard & Indicator-Based Reporting
Data Access Layer
Data Analytics and Reporting
Data Transformation Engine
Staging & Validation Zone

The logical system architecture is divided into six following layers.

1. **User Interface Layer (Presentation Layer):** This layer is responsible for interacting with the user by showing the website. The presentation layer then intern interacts with the business logic layer. The presentation layer includes the Platform interface.
2. **Service Orchestration Layer:** This layer manages how requests and data flow between users and backend systems.
   i. **Service Routing:** Handles the flow of user requests to backend modules, ensuring that queries (e.g., analytics, data upload) are directed to appropriate microservices. It maintains scalability and prevents performance bottlenecks in high-load scenarios.
   ii. **API Gateway:** Facilitates seamless integration with external systems like Bhuvan, VEDAS, National FLR Portals, and state-level Management Information Systems. It ensures secure, standardized data exchange using RESTful APIs and OGC-compliant protocols.
   iii. **App Routing for Local MIS Systems:** This module links the FLR system to existing state applications (e.g., Forest MIS, Soil & Water Mission dashboards), enabling bidirectional data sync & updates for regional planning and reporting.
3. **Common Access Services – Platform Management:** This layer governs security, access control, system integrity, and system-wide communication mechanisms, ensuring compliance with specified governance and integration requirements.
   i. **Authorization & Authentication:** Implements multi-tier user verification protocols, with OAuth 2.0 standards and token-based access. Supports differentiated login interfaces for national (MoEFCC), state (SFDs), district (RFOs), and community-based users. Data access is governed by role and geographic scope. Ensures secure access to sensitive geospatial and institutional datasets. This emphasizes interoperability with national databases and secure access, both addressed here.
   ii. **Role-Based Access Control (RBAC):** Ensures that users interact only with features and data aligned with their roles. For instance, RFOs can upload field data but not approve reports; State MIS officers can edit and validate; MoEFCC can analyze national rollups. Permissions are adjustable by system administrators. This supports institutional capacity-building and tiered data governance across AFOLU sectors.
   iii. **Logging & Audit Trails:** Tracks all system events such as user logins, data uploads, metadata edits, and report approvals. Maintains secure and timestamped logs for each transaction, ensuring traceability and compliance. Logs are periodically reviewed for anomalies or unauthorized actions. This module supports transparency and audit readiness as required in donor- and government-funded initiatives..
   iv. **Messaging & Notifications:** Enables real-time alerts to users for pending validations, conflict resolution, degraded area detection, or restoration status updates. Alerts can be configured by priority, geographic scope, and user group. Delivered via email, SMS, or in-app messages. This function supports

responsive FLR monitoring and escalations, which helps for system efficiency.

v. **System Integration APIs:** Provides standardized APIs for real-time data exchange with external systems like Bhuvan, VEDAS, state forest MIS, and community dashboards. Ensures compliance with OGC/INSPIRE standards for spatial interoperability. Allows plug-and-play integrations in future (e.g., SEPAL, ROAM). This ensures that the system builds upon existing AFOLU data infrastructures rather than duplicating them.

4. **Microservices Layer:** This layer powers the core functionality of the FLR Monitoring & Evaluation System. Each microservice works independently and communicates with others through APIs, ensuring modularity, scalability, and resilience.

i. **User Management & Authentication:** This module enables secure onboarding and access control for users from MoEFCC, State Forest Departments, field staff, and researchers. It supports two-factor authentication, account verification, and password reset. Role-based privileges are assigned (e.g., read-only for researchers, full access for state officers). System admins can deactivate or suspend users to manage platform security.

ii. **Data Ingestion & Integration:** Enables ingestion of data from multiple sources such as field-based mobile apps, satellite APIs (e.g., Bhuvan, SEPAL, VEDAS), Excel templates, and state MIS systems. This requires building on existing monitoring frameworks, and this module harmonizes diverse formats and periodicities into a single backend stream. It supports scheduled sync, metadata mapping, error logs, and duplication checks. Ensures integration with national reporting platforms. The platform is designed with an API-first architecture to enable seamless data exchange and interoperability with other digital systems maintained by GSRLM and the Government of Gujarat. Key integration capabilities include:

- Sync SHG/VO data with existing livelihood program databases.
- Export data in compatible formats (e.g., JSON, CSV, SHP) for integration with other dashboards.
- Integrate with SMS gateways and IVRS systems already used by state departments.

iii. **Spatial Data Management:** Processes geospatial layers (raster and vector) for restoration polygons, degradation zones, and land use classifications. Uses NDVI, NDWI, EVI, NBR indices derived from Sentinel-2, Landsat, and PlanetScope data. Supports overlays with forest boundaries, working plans, and administrative units. Automates spatial joins, zonal stats, and time-series change detection. Enables preparation of maps for reporting under SDG 15.3 and Bonn Challenge.

iv. **WebGIS Visualization:** Provides users with a map-based interface that includes spatial tools such as drawing, measurement, layer toggling, and exporting map views. Integrates layers from FSI, NRSC, and community-led mapping exercises. Supports analysis through filters (e.g., state-wise, district-wise, vegetation type) and legends for interpreting restoration performance.

Enables real-time visualization of polygon updates and satellite change sand this tool supports decision-making by MoEFCC and SFDs.

v. **Dashboard & Indicator-Based Reporting:** Displays real-time and historical data on FLR interventions through interactive dashboards. Supports bar charts, pie charts, histograms, and tabular data views. Reports on indicators such as forest cover improvement, biomass increase, number of hectares restored, and socio-economic benefits. Enables state and national level comparisons, trend analysis, and export of charts in PDF and Excel formats. Supports adaptive monitoring frameworks.

vi. **AI/ML for Restoration Trends & Early Warning:** Implements predictive analytics to identify likely areas of degradation, low survival rates of plantations, or encroachment threats. Trains models using climate data (rainfall, temperature), remote sensing time-series, and historical restoration success data. Generates heatmaps and risk forecasts to support early intervention.

5. **Data Access & Processing Layer – Backend Pipeline:** This is the engine that handles data quality, transformation, analytics, and final reporting, powering all dashboards and visualization.

i. **Data Transformation Layer:** Takes raw input data (field reports, satellite feeds, Excel sheets) and performs preprocessing. Activities include unit conversion (acres to hectares), coordinate transformation (UTM to WGS84), missing value handling, and standardization of indicator units. This ensures consistency across datasets and prepares them for ingestion.

ii. **Validation & Staging Area:** Implements a dual-validation workflow for ecological, social, and institutional data first at state level and then national. Enables reviewers to check uploads against thresholds (e.g., degraded area > 50 ha) and send them back with comments. Logs reviewer decisions, tracks data status, and generates audit logs.

iii. **Analytics & Reporting Engine:** Generates thematic, geographic, and periodic reports on FLR indicators. Produces outputs for SDG 15.3, NDC progress, LDN reporting, and Bonn Challenge Barometer. Customizable by state, forest division, or restoration method (e.g., agroforestry, assisted natural regeneration). Outputs available in tabular, spatial, and visual formats.

6. **Database Server**: The database server will be PostgreSQL Server that allows us to create enterprise geodatabases that can be linked with portal.

3. Technology Stack

Application will be developed using the following technology.

Module	Technology
Front end (Presentation layer)	JavaScript, Java, jQuery, HTML, CSS, Leaflet.js
Business Logic Layer	PHP /Python (Django)/C#.net/ Node Js /Java Springboot
GIS Tools	Open-Source

Mapping Server	GeoServer
Back End (Database)	SQL Server
Web server	XAMP

Server Configuration: The platform will run on the following platform and versions.

Item	Configuration
Processor	2X Intel® Xeon® Gold 6126 CPU@2.50Ghz
Core	64 cores
Operating System	Ubuntu
RAM	256 GB
Database	MS SQL Server Standard Edition
Space	2 TB (Production) and 2 TB (Backup)

Browsers: The application will run on following browsers.

- Internet Explorer 7.0 or higher
- Google Chrome 25 or higher
- Mozilla Firefox 30 or higher
- Safari 5.0 or higher

4. Data Flow Design
   - Design the flow of data through the system, from data collection to reporting.
   - Define data sources, formats, transformation rules, and storage mechanisms.
   - Ensure data integrity, quality, and traceability.
   - Develop a data dictionary to document data elements, formats, and definitions.
   - Design data validation and error handling procedures.
5. Develop Wireframes
   - Create wireframes to visualize the user interface and layout of the digital tools.
   - Ensure the design is user-friendly, intuitive, and accessible (considering users with different levels of technical expertise).
   - Obtain user feedback on the wireframes and iterate as needed.
   - Develop wireframes for different screen sizes and devices (e.g., desktop, mobile).
   - Incorporate accessibility guidelines (e.g., WCAG) in the design.
6. Develop Prototypes
   - Develop interactive prototypes to demonstrate the functionality of the digital tools.
   - Incorporate user feedback to refine the design and functionality.
   - Validate the prototypes with stakeholders, including SFDs, FSI, and community members.
   - Conduct usability testing of the prototypes.

- Refine the prototypes based on testing results.

Activity 2.2: Develop and Test Digital Tools and Applications

This activity focuses on the actual development, testing, and deployment of the digital tools and applications.

Task 2.2.1: Develop Software Applications

1. Front-end Development
   - Develop the user interface of the applications using appropriate programming languages and frameworks (e.g., React, Angular, Vue.js).
   - Ensure the interface is responsive, cross-platform compatible, and visually appealing.
   - Implement user authentication, authorization, and access control mechanisms.
   - Design and implement data visualization components (e.g., charts, graphs, maps).
   - Develop user interface components for data input, display, and interaction.
2. Back-end Development
   - Develop the server-side logic and database interactions using appropriate technologies (e.g., Node.js, Python/Django, Java/Spring).
   - Implement data processing, analysis, and reporting functionalities.
   - Ensure the system is scalable, secure, and performant.
   - Design and implement the database schema.
   - Develop APIs for data access and manipulation.
3. API Development and Integration
   - Develop APIs to enable integration with existing national and state-level data platforms (e.g., Bhuvan, VEDAS).
   - Ensure data exchange is seamless, secure, and efficient.
   - Adhere to relevant data standards and protocols (e.g., OGC standards).
   - Document the APIs and integration procedures.
   - Implement data synchronization mechanisms.

Task 2.2.2: Conduct Testing and Quality Assurance

1. Unit and Integration Testing
   - Conduct unit tests to verify the functionality of individual components.
   - Perform integration tests to ensure different components work together correctly.
   - Document test cases, results, and bug reports.
   - Automate testing processes where possible.
   - Use version control systems to manage code and track changes.
2. User Acceptance Testing
   - Conduct user acceptance testing (UAT) with SFDs and other stakeholders to validate the system meets their needs.
   - Gather feedback on usability, functionality, and performance.
   - Address user feedback and make necessary adjustments.
   - Develop UAT test plans and scenarios.

- Document user feedback and resolutions.

3. Performance and Security Testing
   - Conduct performance testing to ensure the system can handle the expected load.
   - Perform security testing to identify and address potential vulnerabilities.
   - Ensure the system meets relevant security standards and best practices (e.g., OWASP).
   - Conduct load testing, stress testing, and scalability testing.
   - Perform vulnerability scanning and penetration testing.

Activity 2.3: Deployment, Training, and Capacity Building

This activity focuses on deploying the digital tools, training users, and building the capacity of relevant institutions to use and maintain the systems.

Task 2.3.1: Deploy Digital Tools

1. Prepare Deployment Environment
   - Set up the necessary hardware, software, and network infrastructure for deployment.
   - Configure servers, databases, and other system components.
   - Ensure the environment is secure, stable, and scalable.
   - Develop a deployment plan and checklist.
   - Conduct a pilot deployment in a limited environment.

2. Deploy Applications
   - Install and configure the software applications in the deployment environment.
   - Perform final testing and validation before go-live.
   - Monitor the system during and after deployment to ensure smooth operation.
   - Develop a rollback plan in case of deployment failures.
   - Document the deployment process.

Task 2.3.2: Conduct Training Programs

1. Develop Training Materials
   - Develop training manuals, guides, and tutorials for different user groups.
   - Customize training materials based on user roles and responsibilities.
   - Ensure materials are clear, concise, and easy to understand.
   - Develop training materials in multiple formats (e.g., print, online, video) in English and Gujarati.
   - Incorporate hands-on exercises and case studies.

2. Conduct Training Workshops
   - Organize and conduct training workshops for SFDs, FSI, and other stakeholders.
   - Provide hands-on training on how to use the digital tools and applications.
   - Address user questions and provide ongoing support.
   - Conduct train-the-trainer sessions to build local capacity.

- Evaluate the effectiveness of the training programs.

Task 2.3.3: Provide Ongoing Support and Maintenance

3. Establish a Help Desk
   - Establish a help desk to provide technical support to users.
   - Track and resolve user issues in a timely manner.
   - Gather user feedback to improve the system.
   - Develop a knowledge base of common issues and solutions.
   - Provide support through multiple channels (e.g., phone, email, online chat).

4. Provide System Maintenance
   - Perform regular system maintenance, including bug fixes, updates, and upgrades.
   - Ensure the system remains secure, stable, and performant.
   - Address any technical issues that arise.
   - Monitor system performance and identify areas for improvement.
   - Implement a change management process for system updates.

1.1.4 Work Plan

S. No	Key Deliverables	Months																													
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
1	Baseline report (reviewed & approved) on existing Monitoring Mechanisms			D1																											
2	Report on options & recommendations for MER mechanisms						D2																								
3	Approved design and prototype of tool (for data collection and monitoring)									D3																					
4	Development of FLR monitoring tools and SOPs (for data collection and monitoring)																		D4												
5	Pilot testing of data collection and monitoring tools																						D5								
6	Final report on pilot testing																														D6

1.2 Cooperation

1.2.1 Presentation and interaction between the relevant actors in the contracts are of responsibility

User Group	Responsibilities	Interactions
Field Staff (RFOs, Forest Guards, Patrolling Teams)	• Conduct patrolling, plantation validation, and ground-truthing using mobile applications with offline support. • Collect GPS-tagged observations, notes, and multimedia evidence of site status, threats, or activities. • Validate FLR intervention sites and update real-time field observations.	• Monitor via FLR Dashboards and Geospatial Portals • Use validation reports submitted from field units • Review analytics for national and in0ternational reporting
MoEFCC & National FLR Authorities	• Monitor FLR implementation progress aligned with Bonn Challenge, SDG 15.3, and NDC targets. • Review national summary dashboards & evaluation indicators • Lead institutional coordination, policy framing, and reporting.	• Access national WebGIS and FLR Dashboards with role-based secure credentials. • Generate national-level reports and evidence-based insights for international reporting. • Coordinate with GIZ and consortium on policy alignment and updates.
State Forest Departments (SFDs)	• Lead state-level FLR interventions and integrate outputs into national frameworks. • Review ground-truthing, validation reports, and ecological indicators. • Supervise field teams and implement SOPs for data collection.	• Use state-specific dashboards and tools to assess MER progress. • Coordinate data flows from field to central repository. • Participate in workshops, validation sessions, and peer exchanges.
GIS Analysts and Remote Sensing Experts	• Process multi-source satellite imagery (e.g., Sentinel-2, Landsat, Planet Scope) for NDVI, NDWI, NBR. • Develop geospatial layers and validate ground-truth points. • Support dashboard visualizations and spatial analytics modules.	• Access secure portals to upload processed layers. • Sync geospatial products with centralized WebGIS. • Collaborate with field teams and research partners for model validation.
Technical Partners and Research Institutions	• Co-develop MER indicators and methodologies. • Contribute to peer-reviewed validation and capacity building. • Support training material design and data interpretation.	• Access backend datasets for academic analysis. • Provide expert feedback to refine monitoring tools and protocols.

		• Conduct joint field evaluations and workshops.
GIZ Project Managers and Implementation Partners	• Monitor platform rollout, deliverables & timeline adherence. • Facilitate quality assurance and stakeholder communication. • Ensure coordination across consortium partners.	• Use administrative dashboards to review status reports. • Participate in decision-making for platform enhancements. • Align project progress with output milestones.
System Administrator	• Manage server environments, backups, security protocols, and user roles. • Ensure data standardization, uptime & integration with external systems (e.g., Bhuvan, VEDAS).	• Maintain and configure API access and authentication. • Support user onboarding and resolve system-level technical issues. • Perform updates, security patches, and audit logging.

1.2.2 Strategy for establishing cooperation and then cooperating with the relevant actors

To ensure effective implementation of the FLR monitoring system, the cooperation strategy will be grounded in participatory principles, institutional coordination, and technical validation. Cooperation will be achieved through a structured engagement strategy that aligns with the multi-stakeholder architecture involving central and state forest departments, GIS institutions, research bodies, community-based organizations, and enforcement agencies.

The process will begin with a comprehensive stakeholder mapping and institutional assessment during the inception phase, identifying mandates, data roles, and institutional capacities. Based on this, tailored engagement plans will be formulated for each actor group. Regular multi-level stakeholder consultations and co-design workshops will facilitate expectation setting, capture user needs &validate tools and frameworks under development.

A user-centric design methodology will be applied where stakeholders actively contribute to the design, piloting, and refinement of digital tools and MER protocols. Interactions will be formalized through the establishment of FLR Monitoring Committees at national, state, and district levels, serving as coordination and escalation platforms. These committees will provide governance oversight, approve workflows, and promote cross-sector alignment.

To operationalize cooperation, the project will deploy role-specific training programs for field staff, GIS experts, policy planners, and community monitors. These trainings will be supplemented with e-learning platforms, job aids, and in-field mentorship support. Real-time communication channels including mobile-based alert systems, helpdesks, and feedback dashboards will be used to foster bi-directional coordination, especially between field actors and supervisory staff.

A performance monitoring and feedback loop will be institutionalized through regular review of user analytics, field data consistency checks, and stakeholder satisfaction surveys. Insights gathered will feed into iterative tool development, adaptive management decisions, and institutional learning.

Through this collaborative and adaptive cooperation strategy, the project will build shared ownership, ensure data quality, and institutionalize an integrated FLR monitoring system aligned with national reporting priorities and field-level realities.

1.3 Steering Structure

1.3.1 Approach and procedure for steering the measures with the project partners

The project will adopt a robust and inclusive steering structure that reflects the multi-institutional and multi-level collaboration model. This approach ensures seamless coordination between GIZ, the implementing contractor, national and state forest departments, research institutions, and other relevant stakeholders.

A National FLR Steering Group will be constituted, including representation from GIZ, MoEFCC, State Forest Departments, technical agencies (e.g., FSI, NRSC), and community stakeholders. This group will provide strategic oversight, resolve policy-level bottlenecks, and guide cross-sectoral coordination throughout the implementation.

At the state and operational levels, FLR Monitoring Committees will be activated or established, composed of nodal officers from forest departments, technical experts, and MIS planners. These bodies will ensure ground-level coordination, institutional alignment, and periodic review of implementation status. They will also facilitate technical validation, escalate field-level issues, and ensure stakeholder accountability.

The Team Leader of the contractor will serve as the operational focal point, ensuring effective communication between GIZ and all implementation partners. The Team Leader will oversee workstream progress, monitor adherence to timelines, and supervise expert inputs across MER framework development, tool design, and piloting.

Regular structured reviews will be organized (monthly/quarterly), supported by a reporting system and adaptive planning protocols. These mechanisms will enable proactive risk mitigation, cross-learning, and continuous alignment with India's FLR and land degradation neutrality goals.

1.3.2 Description of contractor's contribution to results monitoring and the associated challenges.

The contractor will play a central role in designing, implementing, and facilitating a comprehensive Results-Based Monitoring System (RbMS). This RbMS will track biophysical, institutional, and socio-economic outcomes of FLR initiatives in alignment with national and international commitments (e.g., Bonn Challenge, SDG 15.3, NDCs). The platform will have a built-in M&E layer to track system performance, user adoption, and operational feedback

Our consulting team will implement a six-step results monitoring approach to track and evaluate project outcomes:

i. Review and Refinement of the Monitoring Framework

- Evaluate existing monitoring and reporting systems across AFOLU sectors.
- Ensure alignment with ToR deliverables and validate the indicator frameworks through multi-stakeholder consultations.
- Integrate FLR-specific indicators from forestry, agriculture, and water sectors, focusing on restoration outcomes and landscape-level impacts.

ii. Milestone Integration and Adaptive Planning

- Define detailed milestones during the inception phase based on agreed-upon work plans.
- Incorporate milestones within the RbMS to enable systematic tracking.

- Allow adaptive refinements based on on-ground progress, feedback from state actors, and monitoring committee reviews.

iii. Definition of Monitoring Indicators and Tools

- Co-develop performance indicators for FLR interventions including forest cover change, biomass recovery, institutional performance, and socio-economic resilience.
- Finalize standardized reporting templates, disaggregation protocols, and metadata formats.
- Align tools and protocols with national digital platforms (e.g., Bhuvan, VEDAS, GHG India Platform).

iv. Field Validation and Data Flow Architecture

- Establish multi-level data collection mechanisms, combining field-based apps with remote sensing and MIS inputs.
- Ensure institutional roles are mapped clearly for data validation and reporting responsibility.
- Integrate field supervision, audit trails, GPS validation & error resolution protocols.

v. Analysis, Visualization, and Performance Feedback

- Utilize digital dashboards and analytics tools to assess real-time and historical performance.
- Conduct periodic validation workshops and review meetings with GIZ, SFDs, and other nodal agencies.
- Provide actionable insights for course correction and evidence-based decision-making.

vi. Institutional Learning and Optimization

- Use findings from monitoring to inform policy adjustments and programmatic improvements.
- Facilitate inter-departmental knowledge exchange to support wider adoption of MER practices.
- Address challenges related to interoperability, data inconsistency, and capacity gaps through training, technical backstopping, and performance incentives.

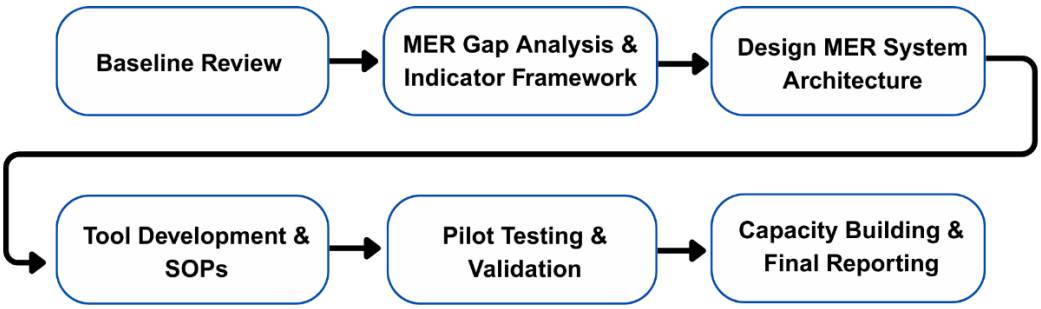
These results monitoring approach is designed to ensure accountability, continuous learning, and strategic alignment with India's restoration goals while fostering cross-sector collaboration and data-driven planning.

1.4 Processes

1.4.1 Presentation and explanation of the implementation plan: work steps, milestones, schedules

The proposed approach for implementing the Forest Landscape Restoration (FLR) Monitoring and Evaluation System will follow a structured, time-bound methodology built around six core work components aligned with the deliverables. The system design will encompass a detailed analysis of current monitoring frameworks, harmonization of indicator definitions, stakeholder alignment, and the digital tool development lifecycle.

Key activities are represented below:



Illustrative Work Plan Flow:

Baseline Review → MER Gap Analysis & Indicator Framework → Design MER System Architecture → Tool Development & SOPs → Pilot Testing & Validation → Capacity Building & Final Reporting

We will draw upon validated datasets, public domain sources, and key inputs from national and state-level stakeholders to inform system design and functionality. The MER system will be co-developed through a participatory process with regular reviews and validation.

Based on this approach, our team will provide ongoing technical and coordination support throughout the 30-month duration of the assignment. Specific details under each work component are described in Section 1.1 (Strategy). Stakeholder consultations will be integrated at each critical stage to validate findings, co-design frameworks, and ensure buy-in. These consultations will involve line departments, technical institutions, civil society, and representatives from MoEFCC and state forest departments. Additionally, focused consultations with implementing partners and domain experts will continue across all stages of the assignment to ensure contextual accuracy and institutional integration.

1.5 Learning and innovation

1.5.1 Contractor's contribution to knowledge management at the partner and at GIZ

Knowledge management is recognized as a key enabler for institutional capacity-building and digital innovation in Forest Landscape Restoration (FLR) and sustainable forest governance. In line with the TOR, knowledge management under this project will serve three core purposes: (i) support learning across forest institutions, (ii) facilitate adaptive management of digital tools and field workflows, and (iii) promote cross-functional integration between geospatial analysis, mobile applications, and operational planning.

The knowledge generated through mobile app development, geospatial dashboard integration, stakeholder consultations, and system testing will be consolidated and systematized to support long-term access, institutional reuse, and inter-agency knowledge exchange within the Government of Gujarat and GIZ.

A centralized digital knowledge repository will be established to consolidate all technical and operational assets developed throughout the project lifecycle. This repository will include:

- A structured library of state forest policies, working plan formats, and standard FLR reporting frameworks relevant to Gujarat's forest management practices.
- Output-based documentation from all work packages, including mobile application workflows, satellite data processing methodologies (e.g., NDVI/NDWI algorithms), user

manuals, SOPs for forest cover change detection, patrolling activity logging, and coupe ground-truthing.

- Categorized stakeholder insights and consultation notes—organized by theme (e.g., enforcement, restoration, planning), geography (district/division), and institutional role (field staff, planners, IT cell).
- Curated technical resources and case references related to digital forest monitoring tools such as Google Earth Engine, QGIS, SEPAL, FLAT, and global best practices on geo-enabled forest management.

To promote accessibility and ensure future usability, the knowledge products will be published in open formats and, where feasible, integrated into GIZ's Atingi platform, the Gujarat Forest Department's digital archive, or shared via national forestry knowledge portals such as ENVIS or ICFRE repositories.

All training sessions led by GISFY will include specific modules on how to navigate, interpret, and contribute to the knowledge repository, empowering forest personnel to continuously build, update, and leverage institutional knowledge beyond the project period.

The project will also promote peer learning and ecosystem knowledge-sharing through technical roundtables, district-level feedback workshops, and state-level review events. These sessions will validate project outputs, collect user feedback, and document field-tested innovations that can inform replication in other states or serve as models for national digital forestry initiatives under RCAP4NDC.

1.5.2 Presentation and explanation of the measures undertaken by the contractor to promote scaling-up effects

To ensure the long-term impact and replicability of the project, GISFY has embedded multiple scaling-up measures throughout its design and implementation strategy. All digital tools mobile applications and the geospatial dashboard are being developed using modular, open-source, and standards-compliant technologies, making them easily adaptable for other forest divisions, states, or sectors such as wildlife conservation or watershed management. The applications will feature configurable parameters and data schemas to allow reuse without extensive redevelopment. GISFY will also document all technical assets, including source codes, APIs, and integration frameworks, in a reusable format to facilitate future customization. Training programs and user manuals will adopt a train-the-trainer model, empowering in-house forest personnel to lead onboarding efforts in other divisions. Additionally, feedback from pilot districts will be codified into a playbook of best practices, challenges, and lessons learned, serving as a reference for future rollouts. GISFY will work closely with the Forest Department and GIZ to align the digital architecture with national-level digital forestry frameworks and encourage its integration into broader FLR and climate resilience programs, thereby laying the groundwork for policy-aligned, state-wide, and even inter-state scaling efforts.

1.6 Project management of the contractor

1.6.1 Approach and procedure for coordination with/in GIZ project

The project management and coordination structure are designed to ensure efficient delivery, timely decision-making, and consistent. A dedicated multidisciplinary team will engage with GIZ's focal point and key national and state stakeholders (e.g., MoEFCC, State Forest

Departments, research institutions) to ensure that activities are contextually relevant, technically sound, and strategically sequenced.

Key coordination mechanisms will include:

- **Inception Meeting:** At the outset of the assignment, a detailed inception meeting will be conducted to clarify project scope, roles and responsibilities, and finalize a mutually agreed communication protocol. This session will also refine the draft work plan and finalize deliverables, timelines, and engagement strategy.
- **Joint Work Plan Development:** During the inception phase, a comprehensive and sequenced work plan will be co-developed with GIZ. It will outline key activities, milestones, responsible experts, and coordination points. The plan will act as the central operational roadmap, guiding delivery and resource allocation throughout the assignment.
- **Structured Progress Reviews:** Regular review meetings scheduled monthly or aligned with key milestones—will serve as checkpoints for tracking implementation progress, addressing challenges, and integrating stakeholder feedback. These reviews will also provide a forum for aligning any course corrections needed due to evolving priorities or emerging insights.
- **Collaborative Reporting Process:** All deliverables will be submitted as per the approved work plan and reporting calendar. Drafts will be shared in advance for structured feedback from GIZ and key stakeholders. The revised versions will integrate this feedback and align with GIZ's Results-Based Management (RBM) framework and quality standards.
- **Adaptive Implementation Approach:** To ensure responsiveness to on-ground realities, a flexible planning framework will be adopted. Insights gained through stakeholder engagement, consultations, or technical analysis will inform iterative updates to activities and deliverables, ensuring continued relevance and effectiveness.
- **Efficient Communication Tools:** Coordination will be supported through shared digital workspaces, version-controlled repositories, and clear assignment of roles for quality assurance, communication, and documentation within the team.
- **Issue Resolution Protocol:** Any operational or coordination challenges will be addressed through a defined escalation mechanism. The Team Leader will serve as the single point of contact for resolving such issues in collaboration with GIZ.
- **Blended Coordination (On-Site and Remote):** While physical meetings and consultations will be prioritized when feasible, the team will also leverage virtual platforms for ongoing coordination, document exchange, and stakeholder engagement ensuring continuity and efficiency.

This coordinated approach ensures that all aspects of the assignment—from analytical review to system design and institutional engagement—are executed collaboratively, transparently, and in alignment with GIZ's objectives and India's FLR goals.

1.6.2 Personnel Assignment Plan (who, when, what work steps) including explanation of and specification of expert months

The following expert deployment plan aligns with the TOR requirements, ensuring adequate coverage across thematic and technical components of the assignment. Travel needs reflect consultations and engagements with state-level actors.

Expert	Expert days in Total	Number of national/ flights (already included in travel budget) (Round Trips)
Team Leader	60	9
Expert 1: Data Scientist	120	9
Expert 2: Ecosystem Modeller	60	9
Expert 3: Geospatial Expert	250	9
Expert 4: Software Engineer	250	9
UI/UX Developer	40	0
Web Developer-1	90	0
Web Developer-2	90	0
Web Developer-3	80	0
Mobile Developers - 1	90	0
Mobile Developers - 2	90	0
QA Engineer	70	0
Deployment Engineer	60	0
Documentation Specialist	40	0

Each expert will be responsible for designated work packages and contribute to cross-cutting tasks such as consultations, quality assurance, and institutional engagement.

1.6.3 Contractor's Backstopping Strategy

The Consultant team understands backstopping as a continuous quality assurance, coordination, and technical support mechanism that ensures high-quality implementation of the assignment. This involves three core dimensions: (a) delivering technically sound and policy-aligned outputs, (b) maintaining project discipline and milestone tracking, and (c) ensuring clear communication and coordination with GIZ and other stakeholders.

Backstopping will be a proactive, structured process that enables real-time support to the core implementation team, particularly during critical phases such as gap analysis, indicator development, stakeholder consultations, and the design of digital tools and governance mechanisms.

Key elements of the backstopping strategy include:

- **Technical Oversight:** The backstopping team will comprise senior experts with domain expertise in FLR monitoring, geospatial systems, institutional governance, and policy analysis. They will validate technical deliverables including monitoring frameworks, indicator sets, gap analysis, and roadmap documents ensuring that they meet both national expectations and international benchmarks such as the Bonn Challenge, SDG 15.3, and LDN.

- **Project Management Support:** The team will support the Team Leader in ensuring that implementation remains on schedule, deliverables meet quality standards, and revisions are efficiently managed. Tools like milestone trackers, Gantt charts, and review protocols will be deployed to enhance project governance.
- **Administrative and Communication Support:** Dedicated personnel will assist in organizing review meetings, coordinating documentation, updating communication channels, and preparing structured progress reports. The backstopping team will also manage version control and internal quality checks before submission to GIZ.

Expert-Based Technical Backstopping Structure

To ensure a comprehensive and impact-driven implementation, the following team of experts will play a key role in technical backstopping and quality assurance:

Expert	Key Responsibilities
Team Leader	Oversee the entire project implementation, ensuring timely delivery of outputs across all work packages. They will coordinate with GIZ, government stakeholders, and consortium partners to manage resources, mitigate risks, and ensure compliance with the TOR. The Team Leader will lead strategic planning, supervise expert teams, and validate quality deliverables. They will also manage reporting, steering committee communication, and stakeholder engagement.
Data Scientist	Lead the design and deployment of analytical models and data pipelines for forest monitoring indicators. They will apply machine learning and statistical techniques to derive insights from field data, satellite imagery, and restoration metrics. The role includes automating data integration, anomaly detection, and trend analysis. The expert will also develop real-time dashboards and support adaptive decision-making through predictive analytics.
Geospatial Expert	Manage spatial database development, satellite image processing, and geospatial layer creation. They will extract forest cover metrics, degradation zones, and restoration suitability using tools like QGIS, ArcGIS, and Sentinel Hub APIs. Their role includes supporting map-based visualization, spatial analytics, and WebGIS integration. They will ensure alignment with national standards and spatial reporting protocols.
Ecosystem Modeller	Develop simulation models to assess forest health, biodiversity, and restoration potential under climate and land-use change scenarios. They will use tools such as InVEST, MaxEnt, and R to model ecological processes and scenario outcomes. The expert will also support the selection of ecological indicators and contribute to decision-support layers. Their work will inform restoration planning and policy alignment.
Software Engineer	Designs and develops the web and mobile applications, backend APIs, and dashboard systems for MER implementation. They will ensure system security, real-time data sync, and performance optimization across platforms. The expert will work closely with GIS and UI/UX teams to ensure seamless user experience. They will also provide technical documentation and long-term maintainability support.

Operational Integration

To ensure the successful implementation of this project, we have assembled a highly skilled multidisciplinary team with expertise in geospatial technologies, forest management, mobile and web development, UI/UX design, quality assurance, and deployment engineering. The team structure is designed to optimize collaboration, efficiency, and innovation.

- The GIS Engineer and Forest Specialist will work closely with the core experts to deliver spatial analysis, indicator validation, and field data integration.
- The Web and Mobile Developers will operationalize tools through scalable applications, dashboards, and automated workflows.
- The UI/UX Designer will ensure accessibility and ease of use across user groups.
- The QA Engineer will implement testing protocols, while the Deployment Engineer will handle staging and rollout of digital components.
- The Documentation Specialist will consolidate technical manuals, user guides, and training materials.

Full CVs and detailed profiles of all team members, including technical and administrative back stoppers, are presented in Section **2 Proposed Staff**.

1.7 Further Requirements

Data Accessibility Support:

The success of this project depends on access to relevant data, field-level reports, working plan documents, satellite archives, and existing digital forest inventory systems maintained by the Forest and Environment Department, Government of Gujarat. Certain critical documents and datasets especially those related to FLR targets, patrolling records, and coupe management may not be readily accessible in the public domain. In order to ensure an accurate and context-aware implementation, we respectfully request the support of GIZ and the Department in facilitating timely access to these internal documents and datasets through appropriate institutional channels. This access will be essential to align system features with on-ground realities, validate digital outputs, and enhance the overall quality, relevance, and performance of the mobile applications and geospatial dashboard developed under the project.

Sustainability Requirements:

Ensuring long-term sustainability is a cornerstone of GISFY's approach to digital platform development. For this project, sustainability will be embedded across technical, operational, financial, and institutional-policy dimensions to ensure that the system remains scalable, adaptive, and serviceable well beyond its initial implementation under RCAP4NDC.

The digital ecosystem including the three mobile applications and the geospatial dashboard will be developed using a modular, scalable, and open-source technology stack to allow future enhancements and integrations. Emphasis will be placed on offline functionality in mobile applications to ensure data collection in forest regions with limited connectivity. Optimization for low-resource environments will include features such as compressed data exchange, caching mechanisms, and lean UI rendering. The architecture will also include robust security protocols, including encrypted data transmission, role-based access control, and compliance with India's Digital Personal Data Protection Act and international standards like GDPR.

GISFY will prioritize building local capacity for long-term operations. A comprehensive knowledge transfer strategy will be implemented through structured training modules tailored for forest officers, technical staff, and data analysts. These will be complemented by easy-to-understand user manuals, SOPs, and video tutorials in English and Gujarati. The system will include a feedback and ticketing mechanism enabling users to report issues and suggest improvements. Regular system audits, performance reviews, and update cycles will ensure that the digital tools remain effective, user-friendly, and aligned with evolving field requirements.

To reduce long-term operational costs, the platform will be hosted on cost-efficient cloud infrastructure (e.g., AWS, Azure, or NIC/GovCloud) with minimal hardware dependency. The system will be designed to integrate with the existing IT infrastructure of the Forest Department, promoting resource sharing and avoiding duplication of services. A maintenance and support budget will be outlined to cover system updates, bug fixes, and user support. GISFY will also advise on potential funding avenues, such as convergence with state digital forestry programs and climate adaptation funds, to support continued operations.

The digital tools developed will be fully institutionalized within the workflows of the Forest and Environment Department, ensuring ownership, accountability, and continuity. The project will be aligned with state-level forest working plans, FLR targets, and national afforestation missions under the Paris Agreement and NDC framework. Furthermore, GISFY will facilitate engagement with academic institutions, NGOs, and knowledge partners to promote inter-agency collaboration and support future expansions, research-driven enhancements, and integration into policy decision-making cycles.

Through this comprehensive sustainability framework, the project will not only enhance the current digital capabilities of the Gujarat Forest Department but also lay a strong foundation for scaling, institutionalizing, and evolving a resilient forest monitoring and restoration system that supports Gujarat's climate and biodiversity commitments in the long run.

2. Proposed Staff

For the successful execution of this project, we have assembled a highly qualified and multidisciplinary team with expertise spanning geospatial analysis, forest management, software development, UI/UX design, quality assurance, and deployment engineering. Our team is structured to ensure that each work package is executed efficiently, with seamless collaboration between domain experts and technology specialists.

Our team's collective expertise and experience in geospatial technologies, environmental conservation, and software engineering uniquely position us to deliver an innovative, data-driven, and impactful forest management solution. Each member has been carefully selected to ensure that all technical, operational, and strategic aspects of the project are executed with high precision and efficiency.

S. No.	Position	Name of the Expert	Task Description
1	Team Leader	Dr. G. Sailu	• Provide overall leadership and strategic direction for project implementation, ensuring alignment with the TOR, RCAP4NDC objectives, and

			Gujarat Forest Department priorities. • Coordinate cross-functional teams involved in mobile app development, geospatial analytics, and stakeholder engagement. • Liaise regularly with GIZ, the Forest and Environment Department, and other stakeholders to review progress, validate deliverables, and incorporate institutional feedback. • Oversee work planning, milestone tracking, and risk mitigation strategies across all project phases. • Ensure the integration of gender equity, sustainability, and policy relevance in all project outputs. • Guide the development of knowledge products, training modules, and institutionalization strategies. • Facilitate review workshops, validation meetings, and final reporting, ensuring project quality, compliance, and knowledge transfer.
2	Data Scientist	Mr. Mohammad Moonis	• Design and implement analytical models for forest cover change detection using satellite imagery (NDVI/NDWI), spatial trend analysis, and real-time data aggregation. • Lead the development of machine learning models to support predictive analytics for deforestation, degradation hotspots, and restoration planning. • Build data pipelines that integrate satellite data, mobile app inputs, and field-verified forest inventory records into a centralized analytical system. • Develop dashboards and visualizations to communicate key performance indicators (KPIs) for forest landscape restoration and patrolling efficiency. • Support the creation of spatial alert systems and automated data-driven decision-support tools. • Contribute to capacity-building by training technical staff on data

			interpretation, dashboard use, and basic analytical workflows.
3	Ecosystem Modeller	Dr. Jayanthi Nagakumar	• Develop spatial ecosystem models to simulate forest dynamics, land-use change, and ecosystem service flows relevant to restoration planning. • Generate vulnerability maps, ecological sensitivity overlays, and suitability zones to guide afforestation and conservation interventions. • Contribute ecological intelligence and modeling outputs to enhance the relevance and functionality of the geospatial dashboard. • Integrate species distribution models and connectivity analyses to support biodiversity conservation planning. • Collaborate with GIS and software teams to embed ecosystem model outputs into the platform's visualization layers. • Support technical workshops by explaining ecological indicators, model assumptions, and how they support decision-making in forest governance.
4	Software Engineer	Mr. Azghar Hussain Mohammed	• Lead the full-stack development of mobile applications for forest cover change monitoring, patrolling activity logging, and working plan coupe ground-truthing. • Develop backend services and secure APIs to ensure smooth data exchange between mobile apps, satellite sources, and the centralized geospatial dashboard. • Ensure mobile applications are optimized for offline use, low-bandwidth conditions, and field-ready usability. • Build and deploy the geospatial dashboard using open-source technologies and integrate it with geospatial servers, satellite imagery, and inventory databases. • Implement system security protocols including data encryption, user authentication, and compliance with national data protection laws.

			• Support testing, debugging, and refinement of digital tools based on pilot feedback and performance evaluation. • Provide technical documentation, system architecture diagrams, and handover protocols for long-term maintenance.
5	UI/UX Designer	Mr. Durga Ajay Babu	• Design user-friendly interfaces for WebGIS, mobile apps, and dashboards, ensuring accessibility for field officers and decision-makers. • Conduct user research and usability testing to refine workflows and interface designs. • Develop interactive wireframes and prototypes to align with project requirements. • Optimize UI/UX for mobile apps to support offline functionality and remote accessibility. • Collaborate with developers to ensure responsive design and seamless user experience.
6	GIS Engineer	Mr. Pranay Kumar	• Oversee geospatial data integration, satellite imagery processing, and spatial analysis. • Develop automated workflows for NDVI, NDWI, and other vegetation indices for deforestation monitoring. • Implement Sentinel Hub API integration for real-time satellite imagery retrieval and processing. • Ensure seamless data synchronization between field-collected GPS data and WebGIS platform. • Support the development of interactive maps, geospatial dashboards, and patrol tracking.
7	Web Developer – 1	Mr. Kishan Kumar	• Develop the frontend interface for the geospatial dashboard and WebGIS platform. • Implement real-time data visualization using GIS libraries (OpenLayers, Leaflet, or Mapbox). • Ensure role-based access controls and interactive analytics features. • Optimize the UI for high-performance rendering of geospatial data.

8	Web Developer – 2	Ms. Pankaja Chichagandi	• Develop server-side APIs for data processing, forest change detection, and integration with Sentinel Hub APIs. • Implement database architecture for storing forest cover changes, patrol data, and GIS layers. • Ensure secure and efficient API connections between mobile apps, GIS servers, and web platforms. • Optimize data processing pipelines for large-scale satellite image analysis.
9	Web Developer – 3	Ms. Girija Priyadarshini Nayak	• Implement secure authentication mechanisms (OAuth 2.0, JWT, role-based access control). • Ensure data encryption and compliance with cybersecurity standards. • Optimize system performance, scalability, and cloud infrastructure. • Conduct load testing and stress testing to enhance platform efficiency.
10	Mobile Developer - 1	Mr. Nirmal k. Shaji	• Develop GPS-enabled mobile applications for patrol tracking and incident reporting. • Implement offline functionality for data collection in remote areas. • Integrate multimedia support (photos, videos, voice notes) for incident reporting. • Ensure seamless data synchronization with the WebGIS dashboard.
11	Mobile Developer - 2	Mr. Mubasheer Ul Islam	• Develop a mobile application for ground-truthing and coupe validation. • Implement GPS navigation features for field officers to locate assigned coupe areas. • Integrate field data validation tools (geo-tagged photos, observations, boundary corrections). • Ensure real-time data transfer and syncing with geospatial databases.
12	QA Engineer	Mr. Mangesh Subash Neul	• Conduct automated and manual testing for web and mobile applications. • Perform unit testing, integration testing, and user acceptance testing (UAT).

			• Identify and resolve bugs, usability issues, and system vulnerabilities. • Ensure cross-platform compatibility and optimized performance.
13	Deployment Engineer	Ms. Fathima Begum	• Set up cloud infrastructure for hosting WebGIS, dashboards, and mobile services. • Oversee deployment, version control & maintenance of the system. • Implement continuous integration and deployment (CI/CD) pipelines for updates. • Monitor server performance, backups & disaster recovery plans.
14	Documentation Specialist	Mr. Siddhartha Goli	• Prepare detailed user manuals, training guides, and technical documentation. • Develop step-by-step guides for mobile app usage, patrol workflows, and GIS dashboards. • Work closely with developers and GIS specialists to ensure accurate documentation of system features. • Create multi-language documentation and e-learning resources for easy adoption by end-users.

2.1 Core Team

The detailed profiles of the experts are described below:

Team Leader: Dr. G.Sailu

Academic Qualification	• Ph. D. in Environmental Science • M.Sc. (Environmental Sciences)
Language	• **English:** Fluent • **Hindi:** Fluent • **Telugu:** Fluent
General Professional Experience	• Over 18 years of experience in managing multidisciplinary projects in the fields of forest governance, climate resilience, and natural resource management. • Proven track record of leading technology-enabled environmental monitoring initiatives, including integration of mobile apps, GIS platforms, and real-time dashboards. • Experienced in project lifecycle management – from inception planning and team coordination to stakeholder reporting and post-deployment support. • Adept at bridging the gap between technical teams, policy institutions, and field stakeholders to ensure effective implementation and adoption of digital tools.
Specific Professional Experience	• Successfully managed projects focused on Forest Landscape Restoration (FLR), afforestation monitoring, and forest inventory systems. • Oversaw development and deployment of mobile-based data collection tools for field forestry teams in India and East Africa. • Led implementation of forest and ecosystem information systems integrated with satellite data (e.g., MODIS, Sentinel-2) and ground-truthing frameworks. • Coordinated capacity-building efforts for forest officers and GIS analysts on restoration tracking, spatial data interpretation, and reporting compliance. • Contributed to the design of climate-smart planning tools for forest and environmental departments, linking field data to policy reporting indicators.
Leadership Experience	• Held lead roles in multiple donor-funded environmental and digital transformation projects, ensuring timely delivery, inter-agency coordination, and policy alignment. • Directed multidisciplinary teams comprising software developers, GIS experts, data scientists, ecologists, and policy advisors. • Served as the primary liaison with government stakeholders, ensuring institutional ownership, policy alignment, and feedback integration.

	• Spearheaded strategic planning and resource allocation, maintaining agile workflows and iterative system development cycles.
Regional Experience	• Extensive implementation experience across India, especially in forest-rich states such as Madhya Pradesh, Maharashtra, Odisha, and Gujarat. • Supported forest governance and ecosystem modeling projects in Nepal and Kenya, contributing to transboundary FLR and REDD+ initiatives. • Familiar with Gujarat's forestry policy frameworks, working plan procedures, and the regional challenges related to afforestation, degradation monitoring, and ecosystem restoration.
Development Cooperation Experience	• Successfully led assignments funded by GIZ, UNDP, FAO, and World Bank focused on digital monitoring systems, environmental governance & sustainable forest management. • Contributed to NDC-aligned programs focused on climate adaptation, land-use governance, and digital innovation under government partnerships. • Well-versed with donor reporting requirements, environmental safeguards, and multi-stakeholder consultation processes.
Others	**Training & Certifications** • Certified in Project Management for Development Professionals (PMD Pro). • Trained in Environmental Monitoring and Evaluation from Wageningen Centre for Development Innovation. • Completed professional courses in Agile Implementation of Digital Tools in Forestry and Land Use Planning. **Technical Expertise & Specialization** • Expertise in integrating digital forestry systems, including mobile data collection, WebGIS dashboards, and spatial analytics platforms. • Skilled in ecosystem-based planning, landscape-level restoration tracking & performance-based results frameworks. • Experienced in using digital tools such as KoBoToolbox, ODK, ArcGIS Online, Power BI, and QGIS for environmental and institutional reporting. • Specializes in institutional sustainability planning, ensuring that digital tools are embedded in long-term departmental workflows and policies.

Data Scientist: Mr. Mohammad Moonis

Academic Qualification	• M. Phil. Natural Resource Management • M. Sc. Environmental Science
Language	• **English:** Fluent • **Hindi:** Fluent • **Korean:** Fluent
General Professional Experience	• Over 17 years of experience in data science, environmental informatics, and statistical modeling applied to climate resilience, forestry, and ecosystem monitoring. • Extensive experience in designing data pipelines, predictive analytics, and AI/ML-based models to support evidence-based decision-making in environmental projects. • Skilled in transforming large, complex datasets (remote sensing, ground data, time-series) into actionable insights for forest planning and restoration tracking. • Expertise in merging structured and unstructured data for real-time monitoring and performance measurement in forest and natural resource systems.
Specific Professional Experience	• Developed machine learning models for forest cover change detection, afforestation monitoring, and carbon sequestration estimation using Sentinel-2, MODIS, and Landsat data. • Designed and implemented real-time alert systems and spatial trend analysis modules for degradation and deforestation pattern identification. • Built automated systems for NDVI/NDWI-based forest health assessment, integrated with mobile data and satellite archives. • Contributed to national-level platforms for ecosystem risk analytics, combining climate variables, vegetation indices, and field observations. • Supported dashboard development by embedding analytics engines for visualizing KPIs, performance trends, and multi-temporal spatial change assessments.
Leadership Experience	• Led cross-functional data teams to develop remote sensing data models and statistical forecasting tools for government and donor-led projects. • Mentored junior data analysts and GIS staff on advanced analytics workflows, open-source tools, and geospatial data integration best practices. • Coordinated with software developers and ecosystem modelers to ensure data-driven components were seamlessly integrated into larger platform architectures.
Regional Experience	• Worked extensively in India, including projects in Gujarat, Madhya Pradesh, Odisha, and the Northeast, applying analytics to support regional forest governance.

	• Contributed to regional datasets and restoration planning tools in Nepal, Kenya, and Bangladesh, supporting transboundary FLR and land-use projects. • Familiar with Gujarat's ecological zones and land-use classification systems, supporting region-specific model tuning and scenario analysis.
Development Cooperation Experience	• Contributed to assignments supported by GIZ, UNEP, and the World Bank, particularly in the domains of forest data systems, ecosystem modeling, and landscape monitoring. • Participated in the development of tools aligned with NDCs, REDD+, and Land Degradation Neutrality (LDN) monitoring frameworks. • Supported reporting and analytics development for international initiatives on climate-smart forestry, carbon accounting, and biodiversity conservation.
Others	**Training & Certifications** • Certified in Applied Data Science (IBM via Coursera). • Completed training in Machine Learning for Remote Sensing (NASA ARSET Program). • Certified in Geospatial AI for Natural Resource Management by the Indian Institute of Remote Sensing (IIRS). **Technical Expertise & Specialization** • Expert in Python, R, Google Earth Engine, SQL, TensorFlow, and Tableau/Power BI for geospatial analytics and environmental modeling. • Specialized in time-series analysis, anomaly detection, supervised classification, and spatial econometrics. • Proficient in integrating satellite-derived indices (e.g., NDVI, EVI, NDWI) with mobile application inputs for real-time monitoring systems. • Skilled in automated reporting systems, model versioning, and deployment of scalable analytical models in cloud environments (AWS/GCP).

Ecosystem Modeller: Dr. Jayanthi Nagakumar

Academic Qualification	• P.hD (Environmental Science) • M.Sc (Biotechnology)
Language	• **English:** Fluent • **Hindi:** Fluent • **Telugu:** Fluent
General Professional Experience	• Over 18 years of professional experience in ecosystem modeling, landscape ecology, and environmental simulation

	applied to forest planning, biodiversity conservation, and restoration assessment. • Extensive experience in using geospatial and ecological datasets to develop landscape-scale models for decision support in forestry and land-use management. • Skilled in linking ecological processes with spatial modeling tools to assess degradation, simulate land-use changes, and plan conservation strategies.
Specific Professional Experience	• Developed ecosystem service models, restoration suitability maps, and landscape vulnerability assessments using tools like InVEST, MaxEnt, and ArcGIS ModelBuilder. • Designed spatial prioritization models for afforestation, forest fire risk mapping, and eco-sensitive zone delineation across diverse ecological regions. • Applied species distribution modeling (SDM) and connectivity analysis to support habitat protection, assisted regeneration, and green infrastructure planning. • Contributed to forest and ecosystem information systems by integrating biotic and abiotic datasets, topographical layers, and anthropogenic pressure indices. • Supported digital platforms with ecological overlays and simulation outputs to inform restoration interventions and scenario-based planning in forest divisions.
Leadership Experience	• Led ecosystem modeling components of multi-institutional projects involving ecologists, GIS experts, and software developers. • Supervised field teams and remote sensing specialists in the calibration and validation of ecological models across diverse landscapes. • Provided technical leadership in stakeholder workshops for ecological zoning, scenario testing, and restoration mapping.
Regional Experience	• Worked on forest ecosystems in Western Ghats, Eastern Himalayas, Central India, and Gujarat, with strong knowledge of dry deciduous and semi-arid forest systems. • Supported ecosystem restoration and habitat connectivity projects in Gujarat and Maharashtra with emphasis on FLR prioritization and biodiversity co-benefits. • Familiar with the ecological diversity, land degradation patterns, and climate vulnerabilities specific to Gujarat's forest zones, enabling effective regional model customization.
Development Cooperation Experience	• Worked on projects funded by GIZ, UNDP, and ICRAF, contributing ecological intelligence to land restoration frameworks, climate adaptation tools & afforestation strategies.

	• Aligned ecosystem modeling efforts with global restoration initiatives such as the Bonn Challenge, REDD+, and UNCCD's Land Degradation Neutrality (LDN) indicators. • Supported the development of ecosystem service-based planning tools integrated into digital forestry platforms under bilateral and multilateral donor programs.
Others	**Training & Certifications** • Certified in Ecosystem Services Modeling using InVEST (Stanford University). • Trained in Climate Change Adaptation and Landscape Ecology from GIZ-AIZ. • Participated in workshops on Remote Sensing for Biodiversity and Habitat Modeling (ISRO-National Remote Sensing Centre). **Technical Expertise & Specialization** • Proficient in tools such as InVEST, MaxEnt, GAMS, R, ArcGIS, QGIS, GRASS GIS, and TerrSet for ecological simulations and spatial analysis. • Specialized in restoration ecology, carbon sequestration modeling, ecological zoning, and integration of ecosystem indicators into spatial dashboards. • Experienced in translating scientific modeling outputs into GIS layers, scenario planning tools, and spatial policy recommendations. • Skilled in preparing vulnerability and suitability maps, ecological impact assessments, and geospatial storytelling for forest decision-makers.

Software Engineer: Mr. Azghar Hussain Mohammed

Academic Qualification	Bachelor's in computer science engineering, Kakatiya University, Andhra Pradesh, India
Language	• **English:** Fluent • **Hindi:** Fluent • **Telugu:** Fluent • **Urdu:** Fluent
General Professional Experience	• Over 22 years of experience in software engineering, enterprise system development, and full-stack application design across sectors including forestry, agriculture, e-governance, and financial inclusion. • Specialized in architecting and delivering scalable, secure, and high-performance digital platforms that integrate geospatial, mobile, and web technologies.

	• Proven ability to lead the end-to-end software development lifecycle, from system design and coding to testing, deployment, and support.
Specific Professional Experience	• Led the development of mobile applications and geospatial dashboards for environmental monitoring, field data collection, and restoration tracking. • Designed and deployed offline-enabled Android apps for field officers to capture patrolling logs, forest cover changes, and working plan coupe validation. • Built secure, API-driven platforms that connect satellite datasets, mobile inputs, and forest inventory databases for real-time analysis and reporting. • Integrated mapping tools (Mapbox, Leaflet) with backend systems to support real-time forest monitoring, alert generation, and visualization of degradation trends. • Developed role-based access systems, data encryption protocols, and compliance layers to ensure secure handling of sensitive environmental and geolocation data.
Leadership Experience	• Managed cross-functional developer teams across large-scale digital transformation projects, coordinating mobile app, web interface, and database development streams. • Oversaw Agile development sprints and DevOps workflows to ensure timely delivery, performance optimization, and cross-platform compatibility. • Mentored junior developers and collaborated with GIS experts, data scientists, and ecological modelers to ensure seamless integration of technical components.
Regional Experience	• Delivered government and donor-funded software projects across India, Africa, and Southeast Asia, including implementations in Gujarat, Maharashtra, Odisha, and Northeast India. • Experience working with state-level forest departments, understanding their digital governance needs and system integration challenges. • Familiar with the forest division structures, working plan systems, and regional technical capacity levels within Gujarat.
Development Cooperation Experience	• Contributed to assignments funded by GIZ, World Bank, UNDP, and other international development partners focused on climate resilience, ecosystem governance, and digital innovation. • Designed secure and scalable software systems for donor-supported platforms in forest inventory digitization, climate information systems, and natural resource governance.

	• Experience working with development cooperation frameworks that require open standards, localization, sustainability, and knowledge transfer.
Others	**Training & Certifications** • Certified in Full-Stack Web Development & Secure API Design (Microsoft). • Completed professional courses in DevOps Automation, Cybersecurity for Cloud Platforms, and Agile Project Management. • Trained in GIS-Application Integration for Forestry and Land Use Monitoring. **Technical Expertise & Specialization** • Proficient in Microsoft ASP.NET, PHP, Python, Angular, React, C#, and frameworks like Laravel and Node.js. • Skilled in geospatial integration using GeoServer, PostGIS, Leaflet, Mapbox, and web mapping technologies. • Extensive experience with mobile-first design, offline capabilities, cloud hosting (AWS, Azure), and database optimization using PostgreSQL, MongoDB, MySQL. • Specialized in RESTful APIs, system security, and data synchronization for real-time, field-to-dashboard communication. • Experienced in developing automated reporting tools, ticketing systems, and post-deployment support protocols tailored for public sector adoption.

UI/UX Designer: Durga Ajay Babu

Academic Qualification	Bachelor of Technology in Electronics and Communication Engineering, Mallareddy Institute of Engineering and Technology, Jawaharlal Nehru Technological University, Hyderabad, India, 2019
Language	• **English:** Fluent • **Hindi:** Good • **Telugu:** Fluent
General Professional Experience	• More than 5 years of experience in UI/UX design, user research, and front-end development. • Specialized in user-centric design, wireframing, prototyping, and data-driven UX improvements. • Strong expertise in GIS-based UI design, working with ArcGIS tools and geospatial platforms. • Experience in designing responsive web applications, dashboards, and mobile interfaces. • Proficient in visual storytelling and branding, with hands-on experience in print and digital design.

Specific Professional Experience	• Lead UI/UX Designer for web and mobile platforms, ensuring a seamless user experience for GIS-based applications. • Designed & Developed interactive UI/UX solutions for ArcGIS-powered platforms, incorporating geospatial data visualization. • Created wireframes, prototypes, and user flows for fintech, healthcare, and e-commerce applications. • Developed mobile-friendly UI frameworks, ensuring accessibility across multiple devices and platforms. • Implemented modern design standards using Figma, Adobe XD, and JavaScript-based UI frameworks.
Leadership Experience	• UI/UX Lead at GISFY Private Limited, overseeing user experience research and front-end design. • Associate UI/UX Designer at IST Private Limited, developing web and mobile UI components for global clients. • Senior UI/UX Designer at REDCON Private Limited, focusing on enterprise application design and usability testing. • Trainer & Mentor for junior designers, conducting workshops on user research methodologies and front-end best practices.
Regional Experience	• Extensive work experience in India, UAE, and the USA. • Designed user-centric web and mobile interfaces for government agencies, private companies, and GIS-based enterprises. • Worked on UX strategies for multinational teams, ensuring localization and accessibility compliance.
Development Cooperation Experience	• Designed GIS-based UI solutions for digital platforms, supporting smart urban planning and environmental mapping projects. • Contributed to UI/UX frameworks for development projects, enhancing data visualization and interactive user experiences. • Assisted in the development of user training materials and documentation for large-scale software deployments.
Others	**Training** • Conducted UI/UX training workshops for corporate clients and internal teams. • Led design thinking sessions, focusing on user behavior analysis, UI trends, and prototyping best practices. **Publications & Certifications:** • Certified UI/UX Designer with expertise in Figma, Adobe XD, and ArcGIS Experience Builder. • Published articles on user research methodologies, UI frameworks, and accessibility in GIS applications. • Contributor to design forums and UX case studies, sharing knowledge on interactive UI development.

	Technical Expertise & Specialization • UI/UX Design Tools: Adobe Photoshop, Illustrator, XD, Figma • Front-End Development: HTML, CSS, Bootstrap, JavaScript • GIS-Based UI Design: ArcGIS API, Web AppBuilder, Experience Builder • UX Research & Testing: User flow creation, wireframing, usability testing.

Geospatial Expert: Mr. Pranay Kumar

Academic Qualification	• Master's Diploma in GIS & Remote Sensing – Jadavpur University, Kolkata (2010) • Master's in Geography – Patna University, Bihar (2003-2006) • Bachelor's in Geography – Patna University, Bihar (2000-2003)
Language	• **English –** Good • **Hindi –** Good
General Professional Experience	• Over 13 years of experience in GIS, remote sensing, geospatial database management, climate change modeling, and spatial analysis. • Expertise in geoportal development, GIS-based decision support systems, and multi-hazard vulnerability mapping. • Extensive experience in land degradation assessment, environmental monitoring, and GIS integration for urban planning and infrastructure development. • Strong background in designing GIS applications, spatial analytics, and integrating remote sensing data for environmental sustainability projects.
Specific Professional Experience	• GIS and Remote Sensing for Climate and Environmental Analysis – Led the development of GIS-based climate resilience platforms, mapping hazard vulnerability, land degradation, and deforestation patterns. • Geoportal & Decision-Support System Development – Designed geospatial databases and web-based GIS platforms for disaster risk reduction, environmental monitoring, and sustainable development planning. • GIS-Based Infrastructure & Urban Development – Conducted spatial analysis, land use planning, and transportation network mapping for government agencies and international development organizations. • Biodiversity & Natural Resource Management – Applied GIS in ecological assessments, watershed conservation planning, and forest landscape restoration.
Leadership Experience	• Served as Project Lead, Programme Manager, and GIS Consultant across multiple organizations, including GISFY

	Private Limited, Development Alternative, and National Informatics Centre (NIC), Govt. of India. • Led multi-disciplinary teams in GIS-based policy advisory, data-driven decision support systems & capacity-building initiatives. • Designed and executed large-scale GIS projects for international donors (GIZ, UNICEF, DFID, World Bank, and Swiss Development Cooperation).
Regional Experience	• Worked on GIS-based projects in India, Malawi, and Afghanistan, applying spatial analysis in land degradation, environmental conservation, and infrastructure planning. • Experience across diverse ecological zones, including Himalayan states, Bundelkhand, Rajasthan, and Bihar, focusing on climate adaptation and natural resource management.
Development Cooperation Experience	• Worked on international development projects funded by the World Bank, GIZ, Swiss Development Cooperation, DFID, and UNICEF. • Ensured GIS-based decision-making for sustainable development, disaster risk reduction and climate resilience planning. • Led GIS-based policy interventions in urban sustainability, forestry conservation, and environmental governance.
Others	**Training** • Conducted GIS training programs for government officials, researchers, and environmental professionals. • Developed capacity-building workshops on GIS application in climate resilience, land use planning, and disaster risk reduction. **Publications & Certifications:** • Published technical reports and GIS-based research studies related to spatial planning, hazard mapping, and geospatial modeling. • Certified ESRI India Lifetime Member, demonstrating advanced proficiency in ArcGIS, remote sensing tools, and geospatial analytics. **Technical Expertise & Specialization** • Forest Cover & Vegetation Mapping – Applied GIS and remote sensing techniques to monitor forest degradation and afforestation patterns. • Land Degradation & Ecosystem Analysis – Conducted spatial analysis for land remediation, biodiversity conservation, and natural resource management.

	• GIS & Decision Support Systems – Expertise in web-based GIS applications, geodatabase design, and geospatial dashboard development. • Disaster Risk Reduction & Climate Resilience – Developed multi-hazard vulnerability indices and GIS-based early warning systems.

Web Developer – 1: Mr. Kishan Kumar

Academic Qualification	Master of Computer Applications (MCA), Dr. Abdul Kalam Technical University, Uttar Pradesh, India
Language	• **English:** Fluent • **Hindi:** Fluent
General Professional Experience	• Over 8 years of experience in software and web development, specializing in full-stack development for enterprise applications, fintech, GIS solutions, and cloud computing. • Strong expertise in Microsoft ASP.NET, PHP, Drupal, AngularJS, ReactJS, Python, Ruby, C#, JavaScript frameworks, and MEAN Stack. • Database Management Expertise: Skilled in MongoDB, MySQL, Oracle, SQL Server, and PostgreSQL for high-performance data handling. • Front-End & Back-End Development: Specialized in API development, web application frameworks, and user interface optimization for large-scale projects. • Experience in GIS-based solutions, financial applications, and agricultural ICT platforms.
Specific Professional Experience	• Full Stack Developer for The Good Platform (Ummeed Vaccination Program): Developed a healthcare monitoring solution integrating mobile applications, GIS mapping, and real-time data visualization. • Software Architect for Lekha – Money Monitoring: Designed and implemented a secure financial tracking system with payment gateway integration, NFC-based smart transactions, and financial analytics dashboards. • Lead Developer for Agri Entrepreneur Growth Foundation (AEGF): Developed a comprehensive ICT platform for agriculture-based entrepreneurship, integrating real-time advisory systems, market linkages, and GIS analytics. • Key Contributor to NTR Smart Township Project: Designed a smart urban planning platform integrating WebGIS, payment gateway solutions, and an automated land allotment system. • Technical Developer for Bharati Makka Application: Built a multi-language agricultural advisory platform, integrating real-time weather updates, pest management insights, and direct farmer-market interactions.

Leadership Experience	• Full Stack Development Lead at GISFY Private Limited, leading web and mobile application development teams. • Key Technology Consultant for multiple fintech and digital transformation initiatives across India, Africa, and Southeast Asia. • Head Developer for Agricultural ICT Platforms, including the implementation of WebGIS, AI-driven advisory, and blockchain-based market linkages. • Trainer and Mentor for junior developers in web technologies, cybersecurity best practices, and financial data security.
Regional Experience	• Extensive experience in India, Africa, Singapore, and the USA. • Implemented ICT-based agriculture platforms in Kenya and Ethiopia, integrating GIS mapping, financial tools, and advisory services. • Developed urban planning technology for Africa and Singapore, incorporating spatial analytics and cloud-based smart city solutions.
Development Cooperation Experience	• Worked with international agencies and development organizations, including: - Asian Development Bank (ADB) – Digital governance and smart city technology integration. - United Nations Development Programme (UNDP) – GIS-based decision support systems for rural economic development. - World Bank & GIZ – Financial technology solutions for inclusive economic growth and secure digital transactions. • Developed ICT-driven agricultural monitoring platforms for government and NGO-led initiatives in South Asia and Africa. • Contributed to climate adaptation digital solutions, integrating real-time monitoring, satellite imaging, and AI-based risk assessments
Others	**Trainings Conducted** • Conducted capacity-building workshops for software developers, GIS analysts, and financial technology experts. • Led trainings on blockchain-based payment solutions, cybersecurity best practices, and cloud computing infrastructure • Designed self-paced learning modules for fintech applications and secure transaction processing. **Publications & Certifications** • Microsoft Developer Group Certification for advanced software development methodologies. • Published technical white papers on GIS-enabled financial systems and cloud-based digital transformation solutions. • Contributor to open-source platforms, including GitHub and CodeProject

	Technical Expertise & Specialization • Programming Languages: ASP.NET MVC, PHP, Angular, React, Python, Java, C#, Node.js. • Database Management: SQL Server, MongoDB, PostgreSQL, Oracle. • Cloud Computing: AWS, Azure, Google Cloud. • GIS Technologies: ArcGIS, GeoServer, WebGIS, Open Layers, Quantum GIS. • Financial Technology & Security: Blockchain, secure payment processing, fintech APIs.

Web Developer – 2: Ms. Pankaja Chichagandi

Academic Qualification	Bachelor of Engineering, Visvesvaraya Technological University, Karnataka, India
Language	• **Kannada:** Fluent • **Telugu:** Fluent • **Hindi:** Fluent • **English:** Fluent
General Professional Experience	• Over 7 years of experience in software and web development, with expertise in full-stack development, UI/UX design, cloud computing, and GIS solutions. • Proficient in C, C++, C#, Java, PHP, Laravel, Drupal, Python, Ruby, WordPress, Leaflet.js, ASP.NET, and JavaScript frameworks (ReactJS, Node.js, Angular). • Database expertise: Skilled in Oracle, SQL Server, MongoDB, MySQL, and PostgreSQL for high-performance data handling. • Specialized in GIS-based development using ArcGIS, QGIS, OpenLayers, and GeoServer. • Experience in developing cloud-based platforms, fintech applications, and e-commerce systems.
Specific Professional Experience	• Lead Developer for Engagement of an IT agency for ICT material development (Agriculture) - ChildFund: Developed an ICT-based agricultural platform featuring project management tools, farmer producer company modules, and an integrated Learning Management System (LMS). • Technical Developer for Agri Entrepreneur Growth Foundation (AEGF): Led the development of a fintech-driven advisory platform providing real-time weather updates, SMS-based farmer advisory, and financial planning tools. • Contributor to TIGRAM - Timber Traceability & Non-Timber Forest Products (NTFP) Management System: Built a forestry governance application integrating QR-based traceability, permit management, and digital market linkages for timber and non-timber products.

Leadership Experience	• Full Stack Developer & Project Coordinator at GISFY Private Limited, leading software development teams for healthcare, agriculture, and fintech projects. • Key Consultant for GIS-based urban planning projects, integrating WebGIS and AI-powered analytics into real estate and smart township development platforms. • Mentor & Trainer for junior developers in cloud computing, API development, and software security best practices
Regional Experience	• Extensive experience in India, Africa, Singapore, and the USA. • Implemented GIS-driven agricultural solutions in India, providing market linkages and decision-support tools for farmers.
Development Cooperation Experience	• Worked with international agencies and development organizations, including: - Asian Development Bank– Digital governance and smart city technology integration. - United Nations Development Programme – GIS-based decision support systems for rural economic development. - World Bank & GIZ – Financial technology solutions for inclusive economic growth and secure digital transactions. • Developed ICT-driven agricultural monitoring platforms for government and NGO-led initiatives in South Asia and Africa. • Contributed to climate adaptation digital solutions, integrating real-time monitoring, satellite imaging, and AI-based risk assessments
Others	**Trainings** • Conducted capacity-building workshops for software developers, GIS analysts, and financial technology experts. • Led trainings on blockchain-based payment solutions, cybersecurity best practices, and cloud computing infrastructure • Designed self-paced learning modules for fintech applications and secure transaction processing. **Publications & Certifications** • Microsoft Developer Group Certification for advanced software development methodologies. • Published technical white papers on GIS-enabled financial systems and cloud-based digital transformation solutions. • Contributor to open-source platforms, including GitHub and CodeProject **Technical Expertise & Specialization** • Programming Languages: C, C++, C#, Java, PHP, Python, Ruby, ASP.NET, Laravel, Drupal, WordPress, ReactJS, AngularJS, Node.js. • Database Management: SQL Server, Oracle, MongoDB, MySQL, PostgreSQL.

	• Cloud Computing: AWS, Azure, Google Cloud. • GIS Technologies: ArcGIS, QGIS, OpenLayers, GeoServer.

Web Developer – 3: Ms. Girija Priyadarsini Nayak

Academic Qualification	Master of Computer Applications (MCA) – Khallikote University, Odisha, India
Language	• **English –** Good • **Hindi –** Excellent
General Professional Experience	• Over 3 years of experience in web development, full-stack programming, database management, and API integration. • Expertise in designing, developing, debugging, and deploying scalable web applications. • Strong background in frontend and backend technologies, specializing in .NET, ReactJS, HTML, CSS, JavaScript, PostgreSQL, and GIS-based applications. • Experience in secure web development, including security audits, content management, and user authentication.
Specific Professional Experience	• GIS and Open-Source Web Development – Proficient in QGIS, PostGIS, GeoServer, OpenLayers, and Leaflet.js, integrating geospatial data into interactive web applications. • Database & API Management – Experience in SQL Server, PostgreSQL, and API integration, ensuring seamless data synchronization and retrieval. • Health & Public Service Web Solutions – Led the development of health monitoring platforms, vaccination management systems, and public service dashboards. • UI/UX & Responsive Web Design – Developed intuitive and user-friendly web interfaces, ensuring cross-browser compatibility and mobile responsiveness.
Leadership Experience	• Held key positions as Web Application Developer at GISFY Private Limited, contributing to high-impact technology solutions. • Led collaborative projects with multiple stakeholders, ensuring seamless implementation of web-based solutions. • Provided mentorship and technical support to junior developers, improving team efficiency and project execution.
Regional Experience	• Extensive work experience in India, with expertise in GIS-integrated solutions for various sectors. • Contributed to state and national-level projects, implementing public health, infrastructure, and environmental monitoring solutions.

Development Cooperation Experience	• Worked on public sector and donor-funded projects, ensuring compliance with government IT and data security standards. • Developed web-based solutions for NGOs and international agencies, supporting data-driven decision-making and policy implementation. • Experience in integrating open-source geospatial tools into public service delivery frameworks.
Others	**Trainings** • Provided technical training and workshops on web development best practices, database management, and API security. • Assisted in training non-technical users on operating GIS-based web applications. **Publications & Certifications** • Certified in ESRI-based GIS development, showcasing expertise in geospatial application development. • Published technical documentation and best practice guides for web application security, deployment, and maintenance. **Technical Expertise & Specialization** • Web Development & Programming – Expertise in ASP.NET, JavaScript, jQuery, ReactJS, HTML, CSS, PostgreSQL, and SQL Server. • Geospatial Web Applications – Proficiency in OpenLayers, GeoServer, PostGIS, QGIS, and Leaflet.js for GIS-enabled interactive web mapping solutions. • Security & Performance Optimization – Implemented role-based access control, authentication mechanisms, and security audits for web-based platforms. • Cloud & Server Deployment – Experience in deploying scalable applications using cloud services, ensuring high availability and performance.

Mobile App Developer – 1: Mr. Nirmal K Shaji

Academic Qualification	Bachelor of Technology in Computer Science & Engineering, Excel Engineering College, Namakkal District, Tamil Nadu, India
Language	• **Malayalam:** Fluent • **Hindi:** Fluent • **English:** Fluent
General Professional Experience	• Over 4 years of experience in mobile application development, specializing in Android and iOS platforms. • Strong expertise in Flutter, Java, C++, C, HTML, CSS, JavaScript, Ionic, Python, PHP, and .NET. • Database Management Expertise: Skilled in Room Database, SQL Server, MongoDB, MySQL, and PostgreSQL.

	• Experience in developing GIS-based mobile solutions, fintech applications, and healthcare platforms. • Worked on government-backed digital transformation projects, focusing on mobile UI/UX, API integration, and cloud-based solutions.
Specific Professional Experience	• Technical Developer for Engagement of an IT agency for ICT material development (Agriculture) - ChildFund: Created an ICT-driven agricultural advisory platform with real-time crop insights, GIS mapping, and mobile-based training modules • Contributor to Agri Entrepreneur Growth Foundation (AEGF): Developed an agriculture-focused fintech application, integrating weather data analytics, SMS-based financial advice, and user-friendly dashboards • Key Mobile Developer for TIGRAM - Timber Traceability & Non-Timber Forest Products (NTFP) Management System: Built a forest governance mobile application with QR-based tracking, digital permit management, and market access tools
Leadership Experience	• Mobile Application Developer at GISFY Private Limited, leading mobile UI/UX and API integration teams for health, agriculture, and financial applications. • Technical Lead for GIS-based Agricultural Solutions, focusing on data-driven insights, weather forecasting, and farmer advisory systems. • Mentor & Trainer for junior developers, specializing in Flutter, API security, and mobile-first design principles.
Regional Experience	• Extensive experience in India, Africa, and the USA. • Developed GIS-based agricultural monitoring platforms for rural areas in Africa, integrating real-time crop advisory and financial tools. • Implemented fintech security solutions for mobile banking applications in the USA. • Worked on urban planning and health-tech mobile applications in India, optimizing mobile accessibility and offline capabilities.
Development Cooperation Experience	• Worked with international development organizations, including: - Takaful – Developed and implemented a mobile application for a crop insurance company, improving user accessibility and operational efficiency. - Designed mobile-based knowledge-sharing platforms for governments and NGOs, advancing financial literacy and providing agricultural training to local communities..
Others	**Trainings** • Conducted mobile app development workshops for government officials, NGOs, and corporate clients.

	• Led training sessions on GIS-integrated mobile applications, fintech security, and API development. • Designed self-paced learning modules for mobile financial solutions and secure online transactions. **Publications & Certifications** • Certified Mobile Developer by Google Android Developer Program. • Published research papers on mobile-based climate risk management and fintech security. **Technical Expertise & Specialization** • Mobile Development Platforms: Android Studio, iOS Development, Ionic, Flutter. • Programming Languages: Java, C++, C, Python, PHP, .NET, HTML, CSS, JavaScript, Ruby. • Database Management: SQL Server, Room Database, MongoDB, MySQL, PostgreSQL. • GIS Technologies: ArcGIS, QGIS, OpenLayers, GeoServer. • Fintech & Security Solutions: Blockchain, NFC transactions, secure payment gateway integration

Mobile App Developer – 2: Mr. Mubasher Ul Islam

Academic Qualification	Bachelor of Technology in Information Technology, Guru Jambheshwar University of Science and Technology, Hisar, Haryana, India
Language	• **Kashmiri:** Fluent • **Urdu:** Fluent • **Hindi:** Fluent • **English:** Fluent
General Professional Experience	• Over 4 years of experience in mobile application development, specializing in Android and iOS platforms. • Strong expertise in Flutter, Kotlin, Java, C++, C, HTML, CSS, JavaScript, Ionic, Python, PHP, and .NET. • Database Management Expertise: Skilled in Room Database, SQL Server, MongoDB, MySQL, and PostgreSQL. • Experience in developing GIS-based mobile solutions, fintech applications, and healthcare platforms. • Worked on government-backed digital transformation projects, focusing on mobile UI/UX, API integration, and cloud-based solutions.
Specific Professional Experience	• Technical Developer for Engagement of an IT agency for ICT material development (Agriculture) - ChildFund: Created an ICT-driven agricultural advisory platform with real-time crop insights, GIS mapping, and mobile-based training modules. • Lead Developer for Lekha – Money Monitoring: Designed a personal and organizational financial management platform with

	secure payment gateways, NFC-based smart transactions, and AI-powered expense tracking. • Contributor to Agri Entrepreneur Growth Foundation (AEGF): Developed an agriculture-focused fintech application, integrating weather data analytics, SMS-based financial advice, and user-friendly dashboards. • Key Mobile Developer for TIGRAM - Timber Traceability & Non-Timber Forest Products (NTFP) Management System: Built a forest governance mobile application with QR-based tracking, digital permit management, and market access tools.
Leadership Experience	• Mobile Application Developer at GISFY Private Limited, leading mobile UI/UX and API integration teams for health, agriculture, and financial applications. • Senior Consultant for Fintech Solutions, implementing secure mobile transaction platforms for financial inclusion projects. • Technical Lead for GIS-based Agricultural Solutions, focusing on data-driven insights, weather forecasting, and farmer advisory systems. • Mentor & Trainer for junior developers, specializing in Flutter, API security, and mobile-first design principles.
Regional Experience	• Extensive experience in India, Africa, and the USA. • Developed GIS-based agricultural monitoring platforms for rural areas in Africa, integrating real-time crop advisory and financial tools. • Implemented fintech security solutions for mobile banking applications in the USA. • Worked on urban planning and health-tech mobile applications in India, optimizing mobile accessibility and offline capabilities.
Development Cooperation Experience	• Worked with international development organizations, including: - Takaful – Developed and implemented a mobile application for a crop insurance company, improving user accessibility and operational efficiency. - Designed mobile-based knowledge-sharing platforms for governments and NGOs, advancing financial literacy and providing agricultural training to local communities..
Others	**Trainings** • Conducted mobile app development workshops for government officials, NGOs, and corporate clients. • Led training sessions on GIS-integrated mobile applications, fintech security, and API development • Designed self-paced learning modules for mobile financial solutions and secure online transactions **Publications & Certifications** • Certified Mobile Developer by Google Android Developer Program

	• Contributor to Hacker rank, Hacker earth, Leet code, publishing insights on mobile UI/UX optimization and GIS-based solutions • Published research papers on mobile-based climate risk management and fintech security **Technical Expertise & Specialization** • Mobile Development Platforms: Android Studio, iOS Development, Ionic, Flutter. • Programming Languages: Java, Kotlin, SQLite, Firebase, MVVM, C++, C, Python, PHP, .NET, HTML, CSS, JavaScript, Ruby. • Database Management: SQL Server, Room Database, MongoDB, MySQL, PostgreSQL. • GIS Technologies: ArcGIS, QGIS, OpenLayers, GeoServer. • Fintech & Security Solutions: Blockchain, NFC transactions, secure payment gateway integration

Quality Assurance Engineer: Mr. Mangesh Subhash Neul

Academic Qualification	Bachelor's in Electronics and Communication Engineering – Kavikulguru Institute of Technology and Science, Ramtek, Nagpur, Maharashtra, India
Language	• **English –** Good • **Hindi –** Excellent
General Professional Experience	• Over 3 years of experience in quality assurance, software testing, and automation testing for web and mobile applications. • Expertise in software development life cycle (SDLC) and software testing life cycle (STLC) to ensure high-quality applications. • Skilled in functional testing, regression testing, performance testing, security testing, and automation testing. • Proficient in manual and automated testing methodologies, ensuring bug-free and optimized applications.
Specific Professional Experience	• Test Planning & Execution – Experience in requirement analysis, test case design, test execution, and defect reporting. • Automated & Manual Testing – Skilled in black-box testing, boundary value analysis, and functional automation testing using Selenium WebDriver and TestNG framework. • Mobile Application & Web Testing – Conducted testing for cross-platform compatibility, performance, and security audits. • QA Documentation & Reporting – Maintained test reports, release readiness assessments, and bug tracking documentation.
Leadership Experience	• Served as QA Engineer at GISFY Private Limited, leading quality assurance initiatives for multiple large-scale projects.

	• Worked collaboratively with developers, designers, and stakeholders to ensure application reliability and compliance with project requirements. • Provided feedback and recommendations for process improvements, enhancing software quality and performance.
Regional Experience	• Work experience in India and Singapore, testing applications tailored for health management, government initiatives, and enterprise solutions. • Expertise in multi-regional compliance testing, ensuring applications meet international data security standards.
Development Cooperation Experience	• Contributed to public sector digital transformation projects, ensuring compliance with government IT policies and best practices. • Worked on healthcare and public service applications funded by government agencies and international organizations. • Developed testing strategies for large-scale, multi-user platforms, ensuring scalability and performance optimization.
Others	**Trainings** • Conducted training sessions for developers and QA teams on software testing methodologies, defect tracking, and performance testing. • Provided workshops on test automation frameworks, helping teams integrate Selenium and TestNG for efficient automated testing. **Publications & Certifications** • Certified in Automated Testing & Quality Assurance, specializing in Selenium, Jenkins, and performance testing tools. • Published technical documentation on testing strategies, automation workflows, and bug tracking best practices. **Technical Expertise & Specialization** • Software Testing & QA Methodologies – Expertise in manual and automated testing for web, mobile, and enterprise applications. • Automation Testing – Experience with Selenium WebDriver, TestNG, Jenkins, and CI/CD integration for automated test execution. • Security & Performance Testing – Conducted vulnerability assessments, load testing, and stress testing for system optimization. • Test Documentation & Compliance – Ensured comprehensive test coverage, defect tracking, and compliance with industry standards.

Deployment Engineer: Ms. Fathima Begum

Academic Qualification	Bachelor of Computer Science Engineering – Vignana Bharathi Institute of Technology, JNTU Hyderabad, India
Language	• **English –** Excellent • **Hindi –** Excellent • **Telugu –** Good
General Professional Experience	• Over 4 years of experience in cloud computing, server deployment, IT infrastructure management, and DevOps. • Expertise in AWS Cloud Resources, EC2 instances, RDS instances, Cloud Watch, Cloud Formation, IAM role management, and automated deployment pipelines. • Strong background in database management, system security audits, and cloud-based application deployment. • Experience in server monitoring, backup management, and disaster recovery planning.
Specific Professional Experience	• Cloud & Server Deployment – Extensive experience in AWS EC2, S3, CloudFront, and Load Balancers (ELB) for scalable and secure system deployment. • Database Management & Archiving – Configured AWS RDS instances, multi-AZ setups, and managed S3 backups for optimized data storage. • Security & Access Management – Created IAM roles, access control policies, and secure server authentication mechanisms. • Software Deployment & Integration – Worked on CI/CD pipelines, automated deployment workflows, and system maintenance strategies.
Leadership Experience	• Served as Deployment Engineer at GISFY Private Limited, ensuring reliable cloud-based solutions for various projects. • Led server migration and cloud transition projects, ensuring minimum downtime and high system availability. • Provided technical guidance to development teams, improving efficiency in cloud computing and IT infrastructure management.
Regional Experience	• Work experience in India and Singapore, specializing in multi-region cloud deployment and IT system optimization. • Experience in configuring cloud-based infrastructure for large-scale applications in public health, environmental monitoring, and enterprise systems.
Development Cooperation Experience	• Worked on government and public-sector IT projects, ensuring secure deployment and regulatory compliance. • Assisted in AWS-based deployment of health management and monitoring platforms, supporting public-sector digital initiatives.

	• Experience in scaling IT infrastructure for international donor-funded projects, ensuring reliability, performance, and security compliance.
Others	**Trainings** • Provided training on cloud computing best practices, server monitoring, and security management. • Assisted development teams in optimizing deployment strategies, reducing downtime, and improving infrastructure performance. **Certifications & Technical Skills** • AWS Certified Cloud Practitioner and Solutions Architect. • Published technical guides and training materials on cloud deployment, security best practices, and IT infrastructure management. **Technical Expertise & Specialization** • Cloud Infrastructure & Deployment – Expertise in AWS EC2, RDS, S3, IAM, CloudFront, ELB, and CloudFormation. • Server & Network Security – Configured secure authentication, firewalls, and encrypted access controls. • Automated Deployment & DevOps – Experience in CI/CD pipelines, Jenkins, and infrastructure as code (IaC) for system automation. • Database & Data Security – Managed SQL databases, AWS backup strategies, and cloud-based disaster recovery plans.

Documentation Specialist: Mr. Siddhartha Goli

Academic Qualification	• Master of Business Administration (MBA) in Systems, Osmania University, Hyderabad, India. • Post Graduate Diploma in Computer Applications, Integrated Institute of Advanced Studies, Hyderabad, India. • Bachelor of Commerce in Computers, Osmania University, Hyderabad, India.
Language	• **English:** Fluent • **Telugu:** Fluent • **Hindi:** Good
General Professional Experience	• Over 4 years of experience in documentation, content management, data entry, and technical documentation. • Expertise in business process documentation, report writing, and content structuring for corporate and educational institutions. • Proficient in Microsoft Office Suite, Advanced Excel, and Adobe Photoshop, with experience in data processing, document design, and structured content management. • Strong knowledge of document formatting, technical writing, and data validation techniques.

	• Experience in working with large-scale corporate documentation and administrative data handling.
Specific Professional Experience	• Documentation Expert for GISFY Private Limited: Leading documentation efforts for software projects, including preparing user manuals, administrative reports, and technical documentation. • Documentation Specialist at Abacus Staffing and Services Pvt Ltd.: Managed technical document creation, compliance reports, and data entry. • Data Entry Operator at Optum Global Solutions: Worked on digital content structuring, records management, and operational documentation. • Computer Operator & Teacher at Sri Chaitanya Techno School: Handled educational material documentation, system records, and computer-based data management. • TIGRAM - Timber Traceability: Developed user manuals and administrative documentation for a forest governance and traceability system. • Agri Entrepreneur Growth Foundation (AEGF): Created business documentation and inception reports for an agriculture-based fintech platform. • Van System: Led documentation efforts for a GIS and data integration system, focusing on workflow mapping and compliance documentation.
Leadership Experience	• Senior Documentation Expert at GISFY Private Limited, leading teams in technical content creation, report structuring, and administrative documentation. • Trainer & Mentor for documentation best practices, ensuring efficient knowledge management and compliance reporting. • Process Improvement Consultant, specializing in streamlining documentation workflows and enhancing digital record-keeping systems.
Regional Experience	• Extensive work experience in India, specializing in documentation and process management for IT and technology-based projects. • Developed software documentation and manuals for government and private sector organizations. • Worked with IT service providers, ensuring technical documentation meets global standards and compliance requirements.
Development Cooperation Experience	• Worked in documentation and compliance structuring for corporate and staffing solutions. • Developed technical documentation for digital transformation projects, ensuring workflow standardization and structured content management.

	• Assisted in business process automation, document digitization, and compliance reporting for corporate and educational institutions.
Others	**Trainings** • Conducted training sessions on document preparation, user manual structuring, and compliance reporting. • Led workshops on Microsoft Office tools, Adobe Photoshop, and Advanced Excel for business documentation. • Trained teams in best practices in documentation management, workflow structuring, and digital content organization. **Certifications & Technical Skills** • Certified in Advanced Excel and Microsoft Office. • Trained in Adobe Photoshop for document design and presentation. • Experienced in data entry automation and structured content development. **Technical Expertise & Specialization** • Software Documentation: User manuals, administrative reports, inception reports, business process documentation. • Content Structuring: Workflow documentation, knowledge management, digital transformation reports. • Data Management: Microsoft Office, Advanced Excel, database entry, structured record-keeping. • Technical Writing & Editing: Compliance documents, training manuals, process improvement reports.

3. Eligibility Requirement of Project Credentials

Our firm possesses extensive experience in geospatial technology, mobile and web application development, satellite imagery processing, and forest management solutions, making us a highly qualified partner for this project. We have successfully implemented GIS-based environmental monitoring platforms, real-time patrol tracking applications, and AI-driven remote sensing solutions in previous projects, ensuring seamless data integration and decision support for stakeholders. Our expertise in Sentinel Hub API integration, NDVI/NDWI-based deforestation analysis, and WebGIS development aligns precisely with the technical requirements of this assignment.

A brief outline of relevant experience, depicting focus areas such as geospatial analytics, mobile data collection, real-time forest monitoring, and government-level IT system integration, is highlighted in the Table below. These projects demonstrate our ability to develop scalable, secure, and user-friendly platforms that enhance decision-making, patrol efficiency, and environmental sustainability. Our team's prior engagements with forest departments, conservation agencies, and international development organizations further validate our competency in delivering high-impact, technology-driven solutions for forest landscape restoration and sustainable resource management.

The details of these citations, methodologies, and technical architecture can be found in the Detailed Credentials Section, where we provide in-depth insights into our past projects, methodologies, and key success factors. By leveraging our extensive experience and deep technical expertise, we are confident in delivering an innovative and scalable digital ecosystem for enhanced forest management in Gujarat.

Annexure "A" – Check list of documents

S. No.	Particular	Document to be attached	References
1	Adherence to GWB Clauses of GAEF	Declaration covering the GWB clauses in Eligibility grid. The details of GWB clause are defined in Legal Inferences document	GWB Declaration under **GAEF Folder**
2	Average Annual Turnover, Financial Statements and Income Tax Compliance (REF: Grid for assessing the eligibility of firms)	• CA Certificate for Average Annual Turnover for the last three financial years bearing UDIN • Audited P&L Account /Income and expenditure sheet and Balance Sheet for last three F.Y along-with the respective Auditor's Reports bearing UDIN Copy of the Recent Income Tax Return submitted or Assessment Order (to compare the turnover as per financial statements with Income Tax Return submitted)	• Annual Turnover Certificate Available under GAEF Folder • Audited financial statements are available under GAEF Folder • IT Returns are available under GAEF Folder
3	Legal Status	Certificate of Incorporation and / other legal registration	Available under **GAEF Folder**

		documents as per the nature of business	
4	PAN, TAN, GST	Copy of PAN, TAN, and GST certificates	Available under **GAEF Folder**
5	Bank details	Copy of Cancelled cheque along with bank details	Available under **GAEF Folder**
6	No. of Employee	Self-declaration or PF statement	Available under **GAEF Folder**
7	Reference Projects	• Work order and/ or Completion certificate/ Experience Certificates of assignments • GIZ Experiences, if any: Please give references of assignments with GIZ. Non-GIZ Experiences: Please give references of assignments other than GIZ.	Available under **GAEF Folder**
8	Technical Experience	• 5 years of experience in designing or developing geospatial web dashboards for planning and reporting purposes. 5 years of experience in providing consulting services for state/national agencies and excellent track record of completion of tasks according to timelines.	Available under **GAEF Folder**
9	Organizational Profile	Organization may submit standard brochure/profile of the organization including the client list for reference purpose	Available under Organization Details (page no. 6)
10	References	Organization may provide reference of minimum 03 reputed clients from last 02 years. Please share the Name, Designation, Name of the organisation, email address or contact number	• **Contact Person:** Mr. Aatish **Designation:** Project Manager **Tel. No.:** +91 8076022747 • **Contact Person:** Dr. Sayaji Hande **Designation:** Project Manager **Tel. No.:** +91 9910510076 • **Contact Person:** Mr. Samuel **Designation:** Project Manager **Tel. No.:** +26772771038
11	Ecological and sustainable parameters	Organization should briefly elaborate the business practices adopted towards the sustainability initiatives and highlight the experience in Sustainable Development Goals (SDGs)	Available under Sustainability requirements under **1.7 Further Requirements**

Annexure 1: Credentials

Assignment No. 01: Agri Entrepreneur Growth Foundation (AEGF)

Assignment name: Agri Entrepreneur Growth Foundation (AEGF)	**Approx. value of the contract (in current US$ or Euro):** USD 50,000
Country: India **Location within country:** India	**Duration of assignment (months):** 11 months
Name of Client: Indus Impact Consulting Private Limited, Mumbai, India	**Total No. of staff-months of the assignment:** 80 months
Contact Person: Mr. Sourabh Gupta **Designation:** Director **Tel. No.: +91 9819432001** **Address:** Aastha, Plot Number 49, Sector-7, Koparkhairne, Navi Mumbai, Maharashtra, 400709 India	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): 16 August 2021	**No of professional staff-months provided by associated Consultants:** 80 Months
Completion date (month/year): 20 June 2022	**Funding Source:** Indus Impact Consulting Private Limited, Mumbai, India
Name of associated Consultants, if any:	
Name of senior professional staff of your firm involved and functions performed (indicate most significant profiles such as Project Director/ Coordinator, Team Leader): **Mr. Azghar Hussain Mohammed – Project Director** Team Management, Documentation (User Manual and Administration Manual), Software Architecture and Application Development and Testing, Conducting workshop	
Narrative description of Project: The Agri Entrepreneur Growth Foundation (AEGF) project was designed to empower rural agri-entrepreneurs through digital solutions that facilitate market linkage, climate-resilient farming, and data-driven decision-making. The goal was to enhance farmers' access to agricultural services, improve productivity, and build a sustainable agribusiness ecosystem using ICT and geospatial tools. The platform leveraged GIS technology to map farmland, assess crop performance using remote sensing, and provide precision farming recommendations. The mobile application developed to connect farmers to markets, provide weather-based advisories, and offer financial services. This platform delivers technology-driven solutions to support data-driven decision-making, streamline field-level operations, and enable dynamic, real-time data visualization.	
Description of actual services provided by your staff within the assignment: 1. Developed a digital platform for Agri Entrepreneurs to manage market transactions, advisory services, and financial records. 2. Integrated GIS-based farm monitoring tools to provide real-time insights on crop health and weather patterns. 3. Designed a mobile application that enabled farmers and agribusinesses to access agricultural advisory services and connect with buyers.	

4. Implemented cloud-based data storage and security mechanisms, ensuring seamless data access and protection.
5. Provided training and technical support to rural entrepreneurs on using the digital tools effectively.

Assignment No. 02: Engagement of an IT agency for ICT material development (Agriculture)

Assignment name: Engagement of an IT agency for ICT material development (Agriculture)	**Approx. value of the contract (in current US$):** USD 30,000
Country: India **Location within country:** India	**Duration of assignment (months):** 16 months
Name of Client: ChildFund, India	**Total No. of staff-months of the assignment:** 80 months
Contact Person: Mr. Santhosh Mishra **Designation:** Project Manager **Tel. No.:** +91 9451796247 **Email:** smishra@childfund.org **Address:** Plot No. 01, 1st Floor, Deen Dayal Upadhaya Marg, New Delhi, 110002, India	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): May 2021	**No of professional staff-months provided by associated Consultants:** 80 Months
Completion date (month/year): Aug 2022	**Funding Source:** ChildFund, India
Name of associated Consultants, if any:	

Narrative description of Project:

This project focused on developing interactive ICT-based learning materials and digital advisory tools for farmers, agricultural extension workers, and policymakers. The objective was to improve knowledge dissemination on sustainable agricultural practices, precision farming, and climate resilience.

GIS technology was integrated into the system to provide spatially-referenced agricultural insights, including real-time weather updates, soil health assessments, and land suitability analysis. The ICT materials were hosted on a web-based GIS dashboard, ensuring that farmers and extension officers could access region-specific agricultural knowledge. The helps in decision-making support through geospatial technology and interactive mobile applications.

Description of actual services provided by your staff within the assignment:

1. Developed an interactive web and mobile platform for farmers to access learning materials, best practices, and agricultural techniques.
2. Integrated GIS-based agricultural maps to support precision farming and crop monitoring.
3. Created multimedia learning content including videos, infographics, and interactive guides.
4. Implemented AI-driven advisory chatbots for automated farmer queries and agriculture best practice recommendations.

5. Designed an offline-compatible mobile application to ensure accessibility in rural and remote areas.

Assignment No. 03: VAN IT System

Assignment name: VAN IT	**Approx. value of the contract (in current US$):** Approx. US$ 37,000 (INR 31,96,677)
Country: India **Location within country:** India	**Duration of assignment (months):** 26 months
Name of Client: Tetra Tech ARD – Forest-PLUS 2.0	**Total No. of staff-months of the assignment:** 110 months
Contact Person: Aatish Mohammed **Designation:** Project Manager **Tel. No.:** +91 80760 22747, +91 92054 09981 **Email:** mohammadaatish@gmail.com **Address:** 3rd Floor, 37 Link Road, Lajpat Nagar III, New Delhi 110024	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): Nov 2022	**No of professional staff-months provided by associated Consultants:** 110 Months
Completion date (month/year): Jan 2025	**Funding Source:** Tetra Tech ARD – Forest-PLUS 2.0
Name of associated Consultants, if any:	

Narrative description of Project:

VAN IT was a comprehensive forest monitoring and biodiversity conservation platform, designed to digitize and track forest-based livelihoods, non-timber forest product (NTFP) collection, and conservation activities. The project played a pivotal role in ensuring sustainable resource management and preventing illegal activities such as poaching and deforestation.

The system incorporated Sentinel Hub API integration for real-time forest cover monitoring, geospatial analytics for tracking patrol routes, and mobile applications for field officers to report illegal activities and NTFP collection data. The platform also provided community-based forest management tools, allowing local stakeholders to engage in afforestation and conservation programs. The system focuses on forest cover change detection, development of interactive geospatial dashboards, and seamless mobile integration to enable real-time field data reporting and validation.

Description of actual services provided by your staff within the assignment:

1. Developed a WebGIS platform to monitor forest areas, patrol routes, and NTFP harvesting activities.
2. Integrated Sentinel Hub APIs for satellite-based forest cover monitoring, enabling real-time tracking of deforestation.
3. Designed a mobile application for forest officers and community members to record NTFP collection, wildlife sightings, and illegal logging activities.

4. Implemented offline functionality, ensuring seamless data collection even in remote forest areas.
5. Developed role-based access controls, allowing forest officials, community leaders, and policymakers to access relevant data.

Assignment No. 04: TIGRAM – Timber Traceability

Assignment name: TIGRAM - Timber Traceability	**Approx. value of the contract (in current US$):** USD 50,000
Country: India **Location within country:** India	Duration of assignment (months): 27 **months**
Name of Client: Tetra Tech ARD – Forest-PLUS 2.0	**Total No. of staff-months of the assignment:** 115 **months**
Name of Client: Tetra Tech ARD – Forest-PLUS 2.0 **Contact Person:** Aatish Mohammed **Designation:** Project Manager **Tel. No.:** +91 80760 22747, +91 92054 09981 **Email:** mohammadaatish@gmail.com **Address:** 3rd Floor, 37 Link Road, Lajpat Nagar III, New Delhi 110024	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): Nov 2022	**No of professional staff-months provided by associated Consultants:** 115 Months
Completion date (month/year): Jan 2025	**Funding Source:** Tetra Tech ARD – Forest-PLUS 2.0
Name of associated Consultants, if any:	

Narrative description of Project:

TIGRAM was developed as a land-use planning and resource management tool, using GIS-based decision support systems to help governments and environmental organizations make data-driven decisions on forest conservation, watershed management, and agricultural land allocation.

This platform utilized Sentinel-2 and Landsat satellite imagery to track deforestation, afforestation, and land degradation trends. It also enabled stakeholders to plan interventions based on real-time spatial analysis. The project included mobile applications for ground-truthing and remote sensing-based classification models to monitor environmental changes.

This application is designed with advanced geospatial dashboards and interactive WebGIS platform, seamlessly integration of Sentinel Hub APIs for real-time satellite imagery processing.

Description of actual services provided by your staff within the assignment:

1. Developed an interactive GIS dashboard for visualizing land-use patterns, deforestation risks, and conservation zones.
2. Integrated Sentinel-2 and Landsat data to enable real-time monitoring of land cover changes.
3. Implemented AI-based classification models to detect forest degradation and land encroachments.
4. Designed a data-driven decision-support system to help policymakers optimize land use and resource allocation.

5. Trained government officials and community organizations on using geospatial tools for land and resource management.

Assignment No. 05: Non-Timber Forest Products

Assignment name: Non-Timber Forest Products MIS	**Approx. value of the contract (in current US$):** USD 45,000
Country: India **Location within country:** India	**Duration of assignment (months):** 6 months
Name of Client: Tetra Tech ARD – Forest-PLUS 2.0	**Total No. of staff-months of the assignment:** 18 months
Name of Client: Tetra Tech ARD – Forest-PLUS 2.0 **Contact Person:** Aatish Mohammed **Designation:** Project Manager **Tel. No.:** +91 80760 22747, +91 92054 09981 **Email:** mohammadaatish@gmail.com **Address:** 3rd Floor, 37 Link Road, Lajpat Nagar III, New Delhi 110024	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): Nov 2022	**No of professional staff-months provided by associated Consultants:** 18 Months
Completion date (month/year): Mar 2023	**Funding Source:** Tetra Tech ARD – Forest-PLUS 2.0
Name of associated Consultants, if any:	

Narrative description of Project:

The NTFP project was aimed at digitizing the monitoring and management of non-timber forest products, ensuring sustainable harvesting, transparent supply chains, and economic benefits for forest-dependent communities.

A GIS-integrated WebGIS platform was developed to track NTFP collection areas, assess ecological impact, and monitor trade activities. Field officers used mobile applications to record NTFP harvest data, enforce sustainable collection limits, and report any illegal harvesting activities. The system also included remote sensing-based vegetation monitoring to assess the availability of NTFPs and predict sustainable harvesting levels.

This focus on data integration, mobile app development for field data collection, and GIS-based decision-making tools for forest management.

Description of actual services provided by your staff within the assignment:

1. Developed a mobile application for forest community members to log NTFP collection data, pricing, and trade activities.
2. Integrated blockchain-based supply chain tracking to enhance transparency in NTFP trading.
3. Designed a WebGIS interface for forest officials to monitor NTFP harvesting areas and ensure compliance with sustainable harvesting guidelines.

4. Implemented remote sensing analytics to assess the impact of NTFP harvesting on forest ecosystems.
5. Developed machine-learning models to predict NTFP availability and optimize harvesting schedules.

Assignment No. 06: Bharati Makka App

Assignment name: Bharati Makka App	**Approx. value of the contract (in current US$):** USD 36,000
Country: India **Location within country:** India	**Duration of assignment (months):** 13 months
Name of Client: Indian Council of Agricultural Research	**Total No. of staff-months of the assignment:** 46 months
Contact Person: Dr. Himanshu Pathak **Designation:** Secretary (DARE) and Director General (ICAR) **Phone:** 23382375; Extn. 6528 **Email:** dirdar.icar@nic.in **Address:** Krishi Bhawan, New Delhi-110 001	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): May 2021	**No of professional staff-months provided by associated Consultants:** 46 Months
Completion date (month/year): Nov 2022	**Funding Source:** Indian Council of Agricultural Research
Name of associated Consultants, if any:	
Narrative description of Project: The Bharati Makka Application was a precision agriculture and market linkage platform for maize farmers, designed to enhance crop productivity, streamline supply chains, and provide market intelligence. The platform integrated GIS and remote sensing to analyze crop growth, detect pest infestations, and optimize irrigation planning. AI-driven advisories provided real-time weather alerts, disease management strategies, and market price forecasts. Farmers could connect directly with buyers through the digital marketplace, ensuring better price realization and reduced post-harvest losses. The project aligns with the TOR's objectives of integrating Sentinel Hub APIs for real-time data processing, providing interactive dashboards for decision support, and leveraging mobile technology for real-time field data collection.	
Description of actual services provided by your staff within the assignment: 1. Developed a mobile application with real-time weather updates, pest alerts, and agronomic advisory services. 2. Integrated geospatial crop monitoring to assess maize growth stages and predict yield potential.	

3. Implemented farmer-to-market linkage features, enabling farmers to connect with buyers and optimize pricing.
4. Designed AI-powered chatbots for automated support on crop diseases, pest management, and best farming practices.
5. Conducted training programs to educate farmers on digital farming techniques and application usage.

Assignment No. 07: Millet Market

Assignment name: Millet Market	**Approx. value of the contract (in current US$):** USD 34,000
Country: India **Location within country:** India	**Duration of assignment (months):** 14 months
Name of Client: Indian Council of Agricultural Research	**Total No. of staff-months of the assignment:** 61 months
Contact Person: Dr. Himanshu Pathak **Designation:** Secretary (DARE) and Director General (ICAR) **Phone:** 23382375; Extn. 6528 **Email:** dirdar.icar@nic.in **Address:** Krishi Bhawan, New Delhi-110 001	**Approx. Value of the service provided by your firm under the contract (in current US$ or Euro):** NA
Start date (month/year): Aug 2021	**No of professional staff-months provided by associated Consultants:** 61 Months
Completion date (month/year): Dec 2022	**Funding Source:** Indian Council of Agricultural Research
Name of associated Consultants, if any:	

Narrative description of Project:

The Millet Market Application was developed to empower millet farmers by digitizing the agricultural value chain, providing real-time pricing data, and enabling direct farmer-to-buyer transactions.

The application featured GIS-based crop production mapping, satellite-derived yield estimation models, and AI-driven demand forecasting. Farmers could access climate-smart agricultural recommendations, storage solutions, and trade opportunities through an interactive digital marketplace.

By integrating GIS-based analytics to map millet production zones and assess supply chain efficiency, the project aligns with the TOR's emphasis on geospatial decision-support tools, real-time data integration, and interactive dashboards for sustainable resource management.

Description of actual services provided by your staff within the assignment:

1. Developed a digital marketplace for millet farmers, enabling direct sales to buyers, processors, and retailers.
2. Integrated AI-driven market price prediction models, helping farmers make informed selling decisions.
3. Designed a mobile app with real-time updates on demand trends, storage solutions, and supply chain logistics.
4. Implemented blockchain-based traceability solutions, ensuring supply chain transparency and quality assurance.
5. Developed GIS-based analytics to map millet production zones and identify optimal market opportunities.